Lecture 3: Multimodal LLMs

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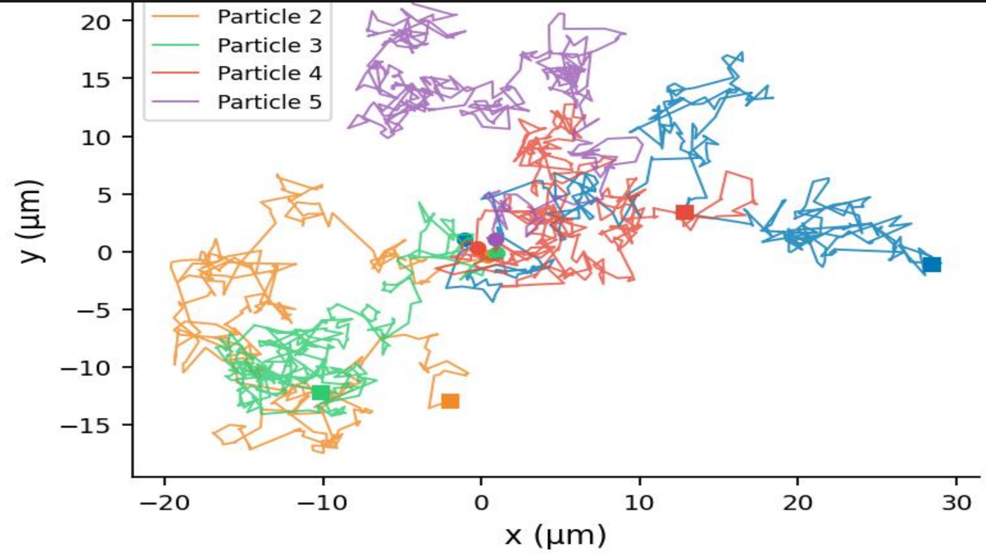
Research:

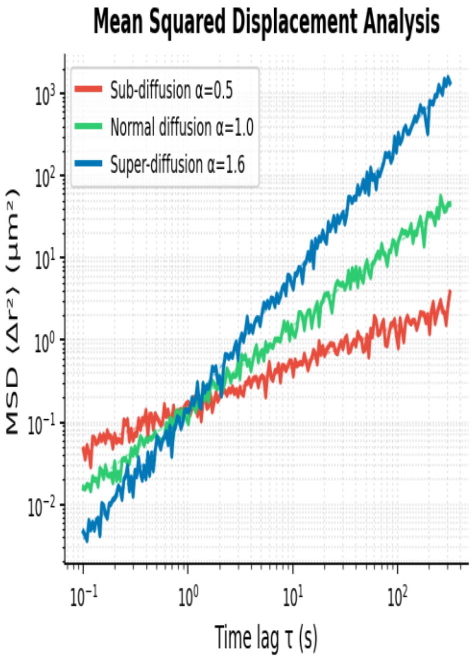
Anomalous diffusion and intracellular transport
Biological physics & stochastic processes

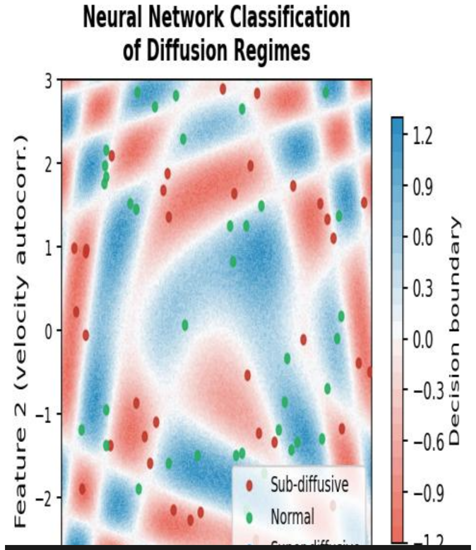
I specialise in:
Stochastic modelling and machine learning of biological applications, Bayesian statistics, single particle tracking analysis

Talk to me about your project ideas!

Anomalous intracellular transport







Machine learning for diffusion classification

Key Applications

Cellular Biology

Modelling movement of vesicles and organelles inside living cells. Single particle tracking and machine learning to classify dynamicalregimes in heterogeneous intracellular environments.

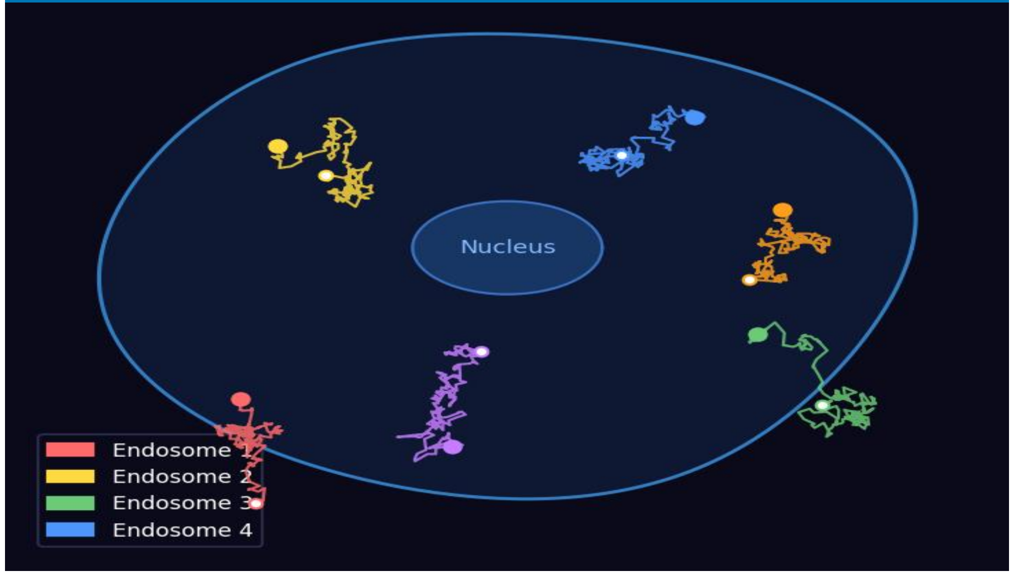
Medicine

Applying mathematical modelling and Bayesian statistics to understand biological processes relevant to disease. Mathematical models describe how cells and molecules behave in complex tissue.

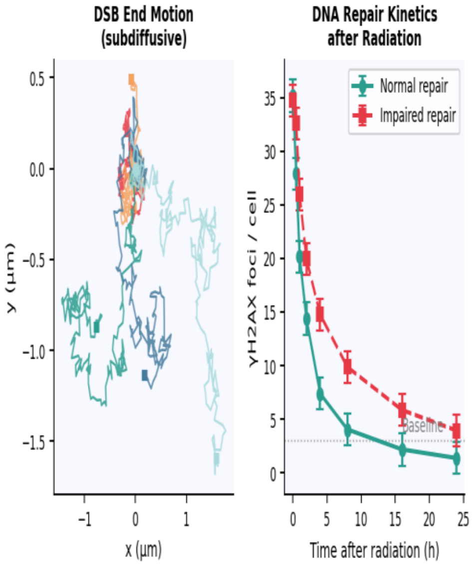
DNA Repair & Radiotherapy Optimisation

Modelling motion of DNA double-strand break ends during non-homologous end-joining repair. Funded by EPSRC in collaboration with The Christie NHS Foundation Trust — the largest cancer centre in Europe — to improve dose planning and outcomes in proton and photon radiotherapy.

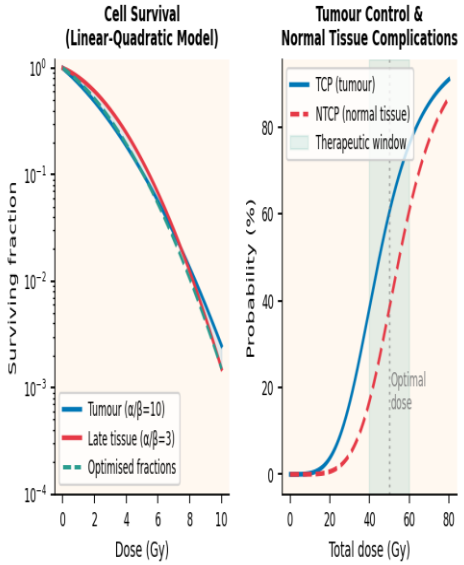
Cellular Biology



DNA Repair



Radiotherapy



Lecture Outline

01	**Text-Only LLMs**	Transformers, attention, pre-training, instruction fine-tuning, RLHF	15 min
02	**Vision Meets Language**	CNNs → ViT, CLIP, Flamingo, BLIP-2, LLaVA — architectures compared	20 min
03	**Image Description with LLMs**	Captioning, VQA, OCR, chart understanding, spatial reasoning, examples	20 min
04	**Llama 3.2 Vision Deep Dive**	ViT-H/14, image tiling, cross-attention, 4-stage training pipeline	20 min
05	**Healthcare Fine-Tuning**	Medical imaging, dataset curation, LoRA fine-tuning, clinical examples	15 min

Part 1: Text-Only LLMs Recap

Re-evaluating Self-Attention, Next-Token Objectives, and Limits of Blind Models

What Is a Large Language Model?

- A language model assigns probabilities to sequences of text tokens
- "Large" = billions of parameters, trained on trillions of tokens
- Core task: predict the next token given all previous tokens (autoregressive)
- No explicit supervision — the data labels itself (self-supervised learning)
- Emergent abilities arise unpredictably at scale: reasoning, coding, translation
- Examples: GPT-4, Llama 3, Mistral 7B, Gemini, Claude 3.5

Core Idea: P(next token | context)

The	cat	sat
on	the	**?**

P("mat" | ...) = 0.34 ✅

P("floor" | ...) = 0.28

P("roof" | ...) = 0.08

Key milestones: GPT-1 (2018) → BERT (2018) → GPT-3 (2020) → InstructGPT (2022) → Llama 3 (2024)

Tokenisation: How Text Becomes Numbers

- Neural networks operate on vectors, not text — tokenisation is the bridge
- Most LLMs use Byte-Pair Encoding (BPE): merges frequent character pairs iteratively
- Vocabulary: typically, 32K–128K tokens; sub-word units handle rare/novel words
- Spaces and punctuation are often fused into tokens
- Numbers are often split digit-by-digit: '2024' → ['20','24'] or ['2','0','2','4']

⚠ Tokenisation affects model behaviour: arithmetic is hard because each digit is a separate token; rare words are split into many pieces; different languages tokenise very unevenly.

BPE Tokenisation Examples

"Hello world"
Tokens: ["Hello"," world"]

"Multimodal"
Tokens: ["Multi","modal"]

"ChatGPT"
Tokens: ["Chat","G","PT"]

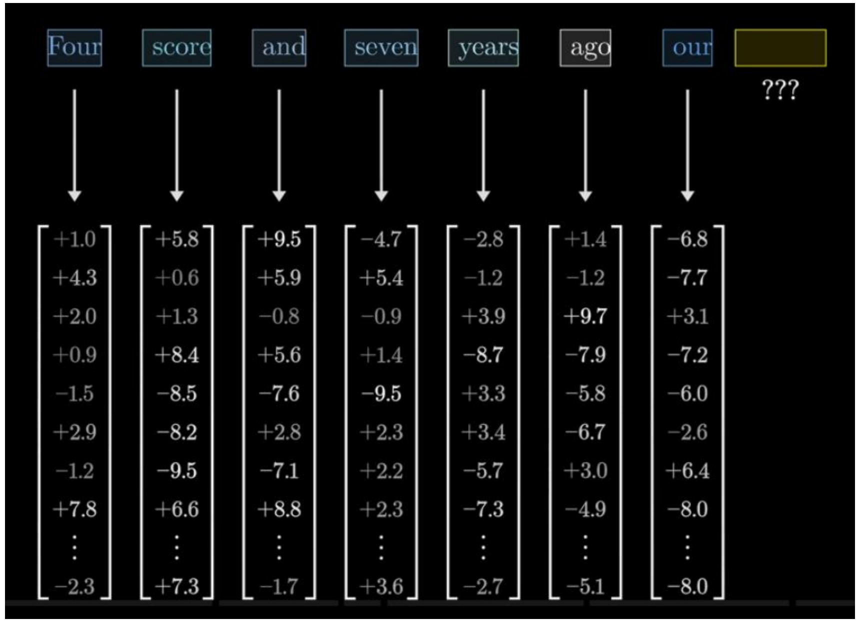
"COVID-19"
Tokens: ["COV","ID","-","19"]

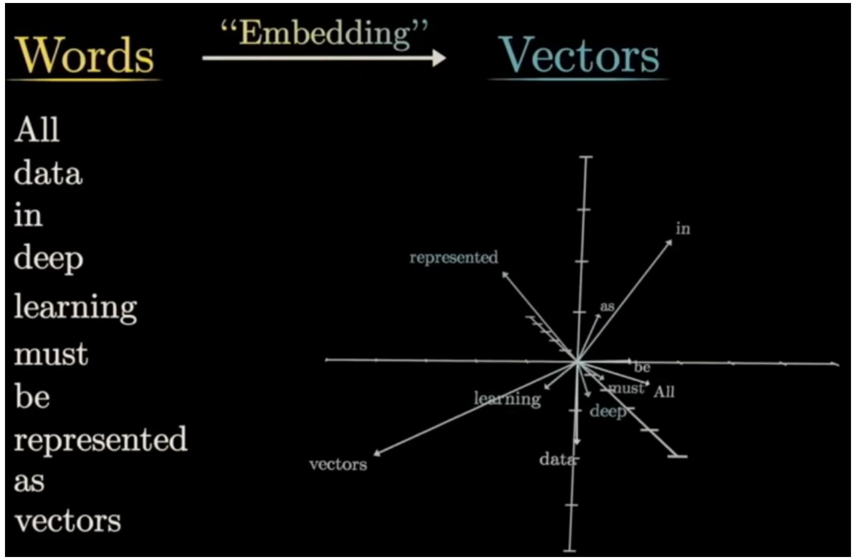
- Associate each word (token) with a (very) high dimensional vector (embedding)

- Directions can correspond to semantic meaning

- Words with similar meaning tend to cluster near each other

3Blue1Brown
Animated math videos, https://www.3blue1brown.com/





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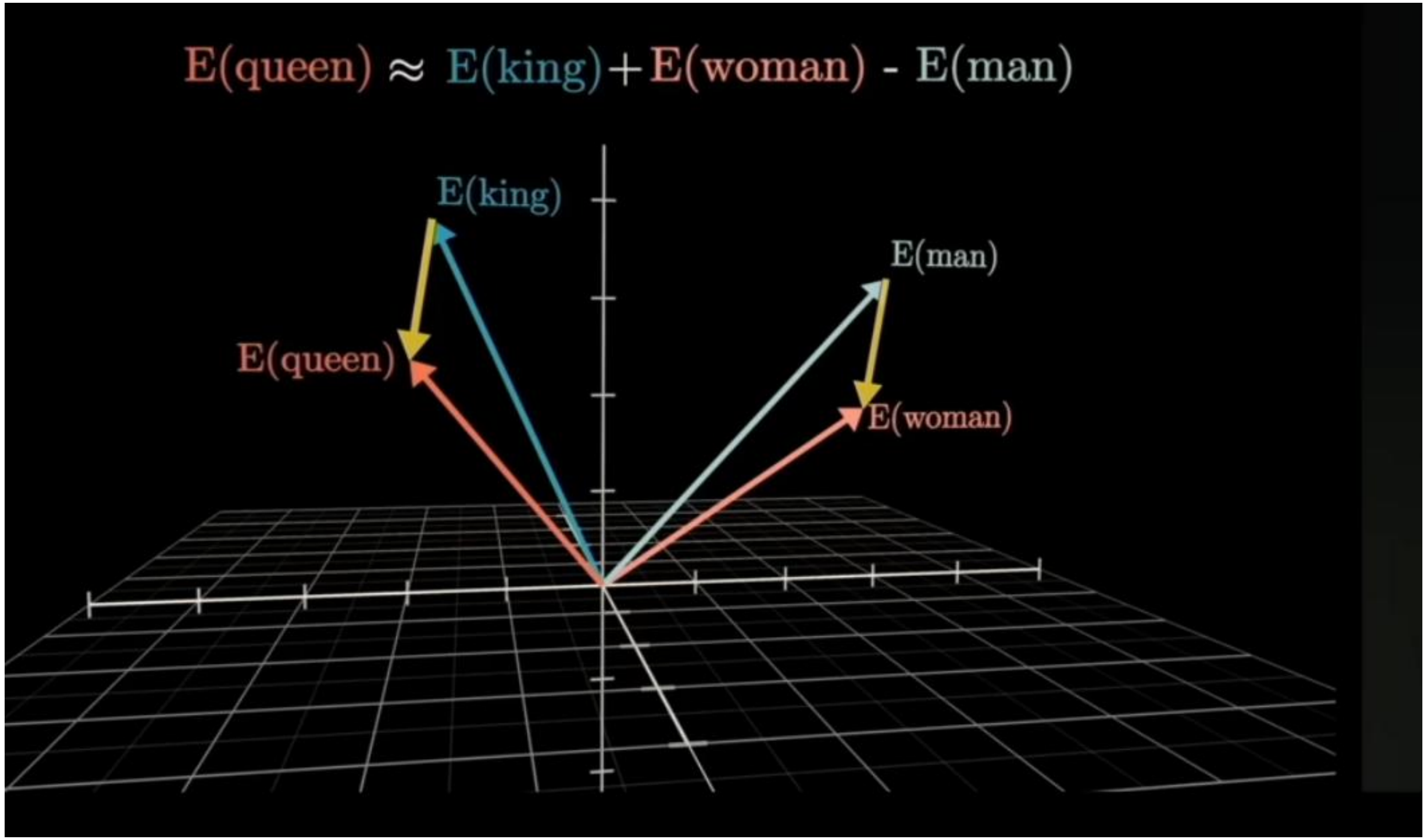
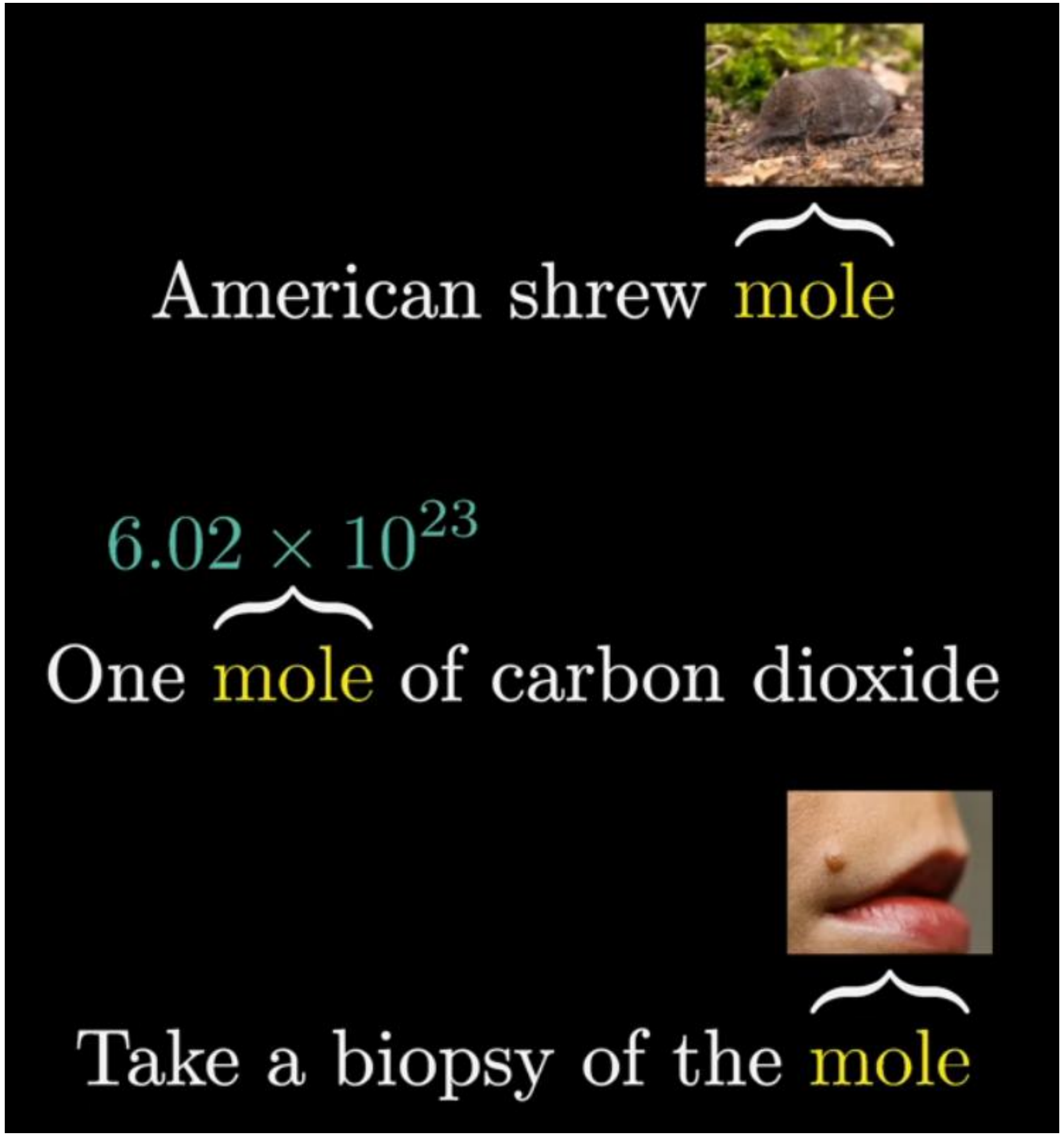
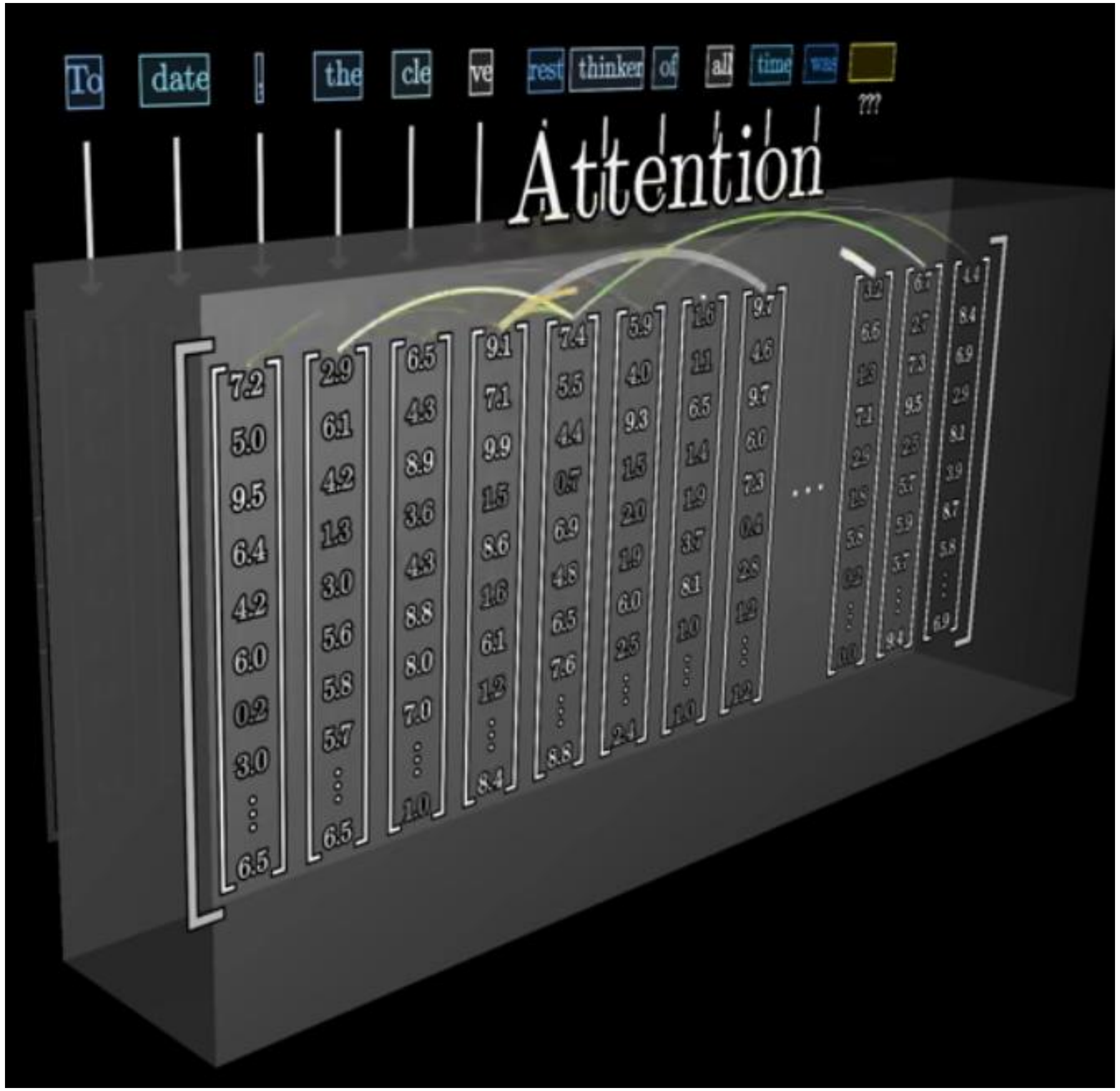


- Why we need attention?

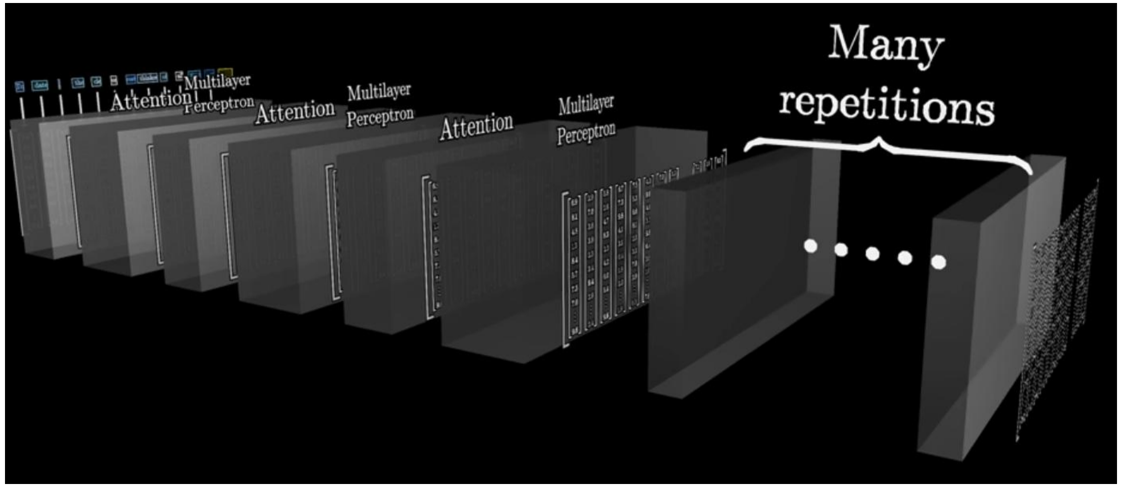
- Vector embedding of token "mole" would be the same in all three sentences

- We need to provide a contextual meaning





- Attention allows to link tokens

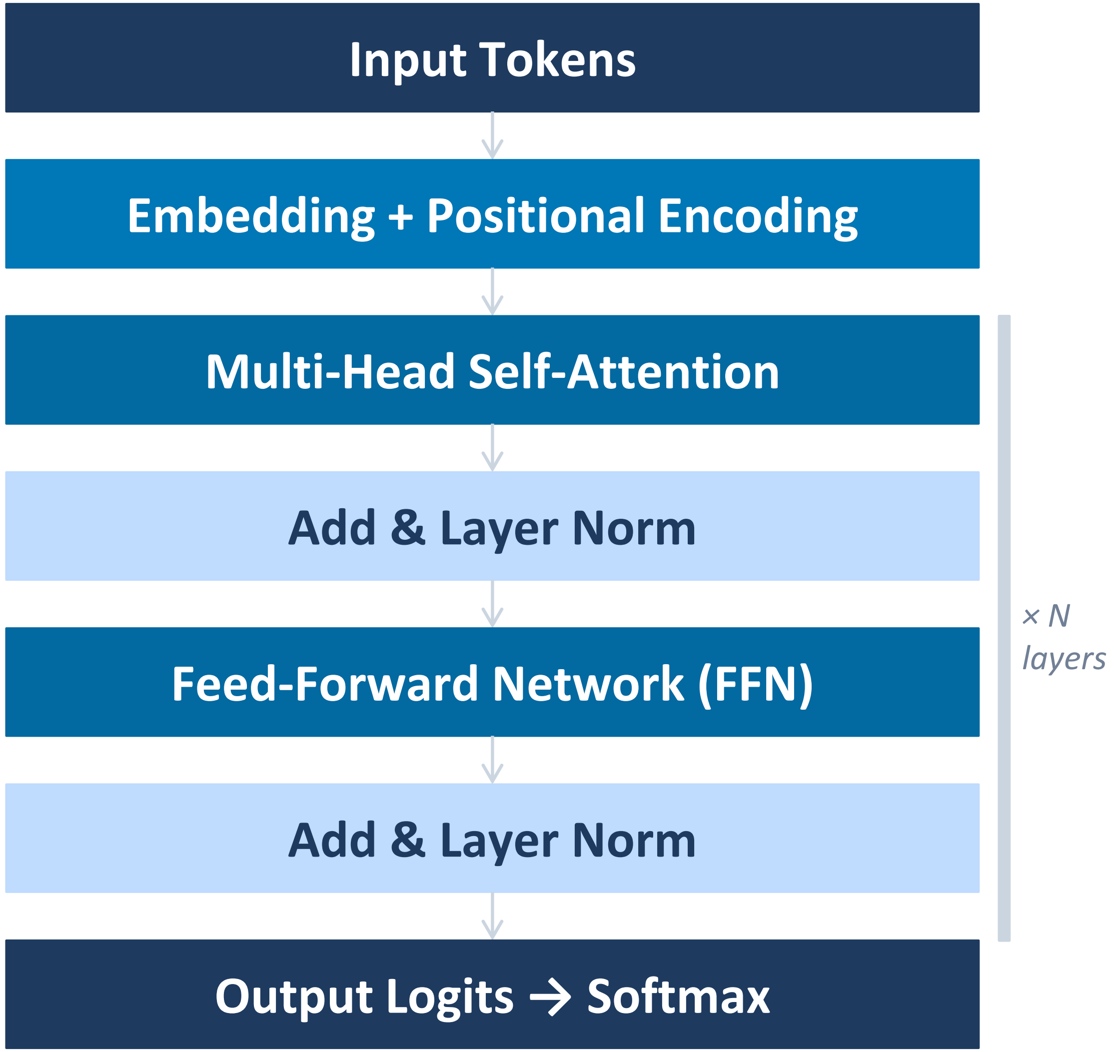


- Transformer architecture (Google, 2017)

The Transformer Architecture (Vaswani et al., 2017)

- Fully parallelisable: no sequential dependency unlike RNNs/LSTMs
- Self-attention captures long-range dependencies
- Multi-head attention = H parallel heads (ChartGPT3 H=96)
- Positional encodings inject sequence order into the representation
- Modern LLMs use decoder-only GPT-style transformers with causal masking

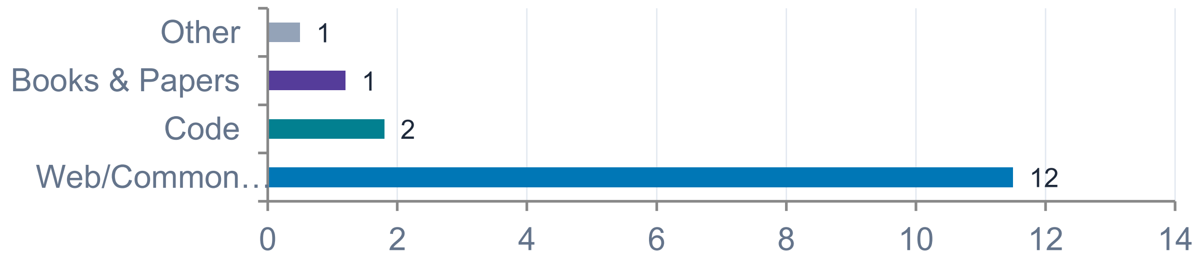
Decoder-only transformers (GPT, Llama) attend to all previous tokens via a causal (lower-triangular) mask. Each token can only see itself and the tokens before it.



Pre-Training: Next-Token Prediction at Scale

- Training data: trillions of tokens — Wikipedia, books, code, arXiv…
- Objective: given context ["The","cat","sat"], maximise P("on")
- No human labels needed — the data labels itself (self-supervised)
- By treating the data as both the input and the target, the model generates its own supervisory signals, eliminating the need for expensive human labelling.

Llama 3.1 405B pre-training data (~15T tokens)



Training Step: Next-Token Prediction

The	quick	brown	fox	jumps
over	the	lazy	**dog**	

At each position, model sees previous tokens and predicts the next one.

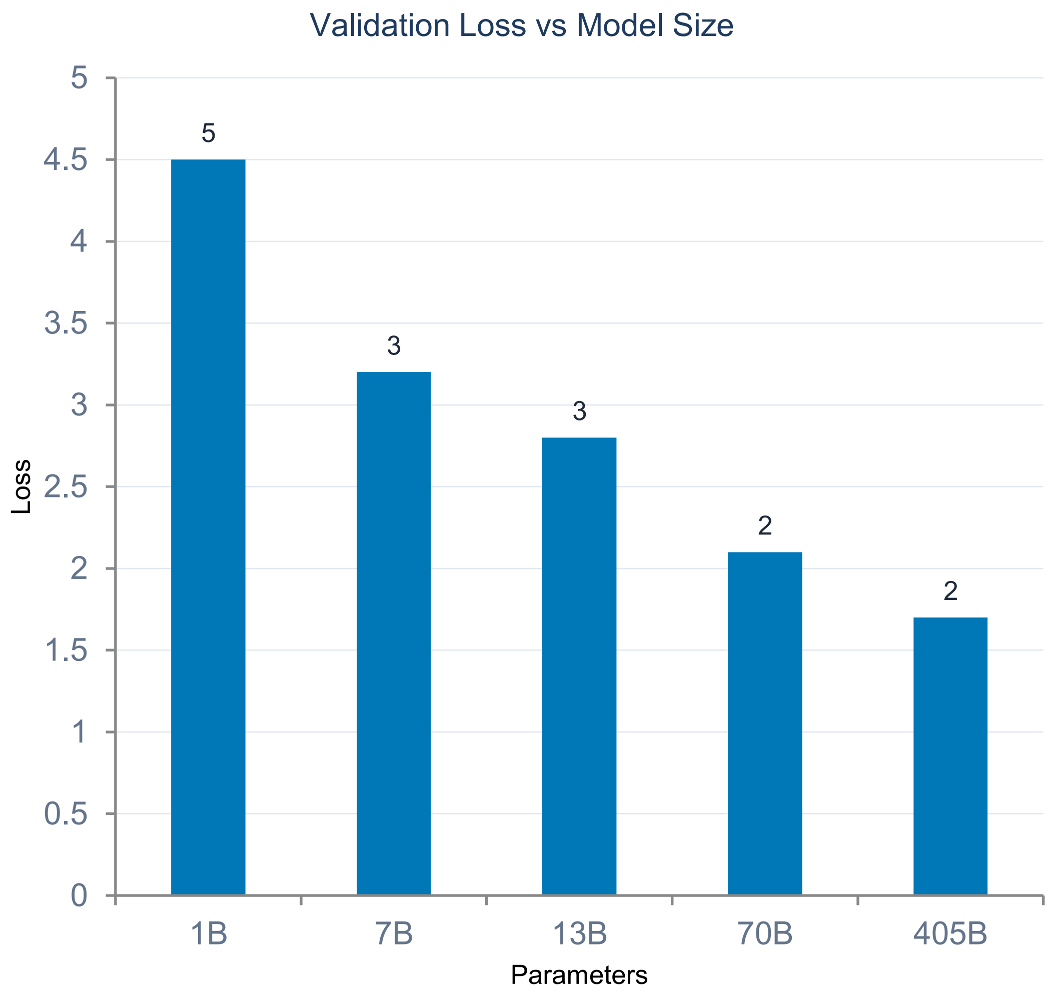
Loss = −log P(dog | The quick brown fox jumps over the lazy)
Summed over all positions → minimise with gradient descent

Scaling Laws and Emergent Abilities

- Kaplan et al. (2020): loss scales as a power law in model size N and data D
- Chinchilla (Hoffmann et al. 2022): optimal to scale N and D equally
- Rule of thumb: ~20 training tokens per parameter
- Llama 3.1 (405B params) trained on ~15T tokens

Emergent Capabilities

- Abilities that appear sharply at certain scale
- Chain-of-thought reasoning, multi-step maths, code synthesis
- Few-shot in-context learning from examples in the prompt
- Cannot be predicted simply by extrapolating from small models



From Pre-Trained Model to Assistant: Fine-Tuning

Step 1: Pre-Training → **Step 2: Supervised Fine-Tuning** → **Step 3: RL**

Step 1: Pre-Training	Step 2: Supervised Fine-Tuning	Step 3: RL
Predict next token on trillions of web tokens. Learns language structure.	Train on curated (instruction, response) pairs. Learns to follow.	Human raters rank outputs. Model learns preferred behaviours and safety.

SFT Training Example

User: Summarise this paper in 3 bullet points: "Attention is All You Need..."

Assistant:

- Introduces the Transformer...
- Uses self-attention instead of RNNs...
- Achieves BLEU 28.4 on EN→DE translation task

Reinforcement Learning from Human Feedback

Prompt: "Explain quantum entanglement simply"

✅ CHOSEN: Clear analogy-based 3-sentence explanation that is accurate and engaging.

✗ REJECTED: Vague jargon-filled answer with no practical example or structure.

Text-Only LLMs: Capabilities and Limitations

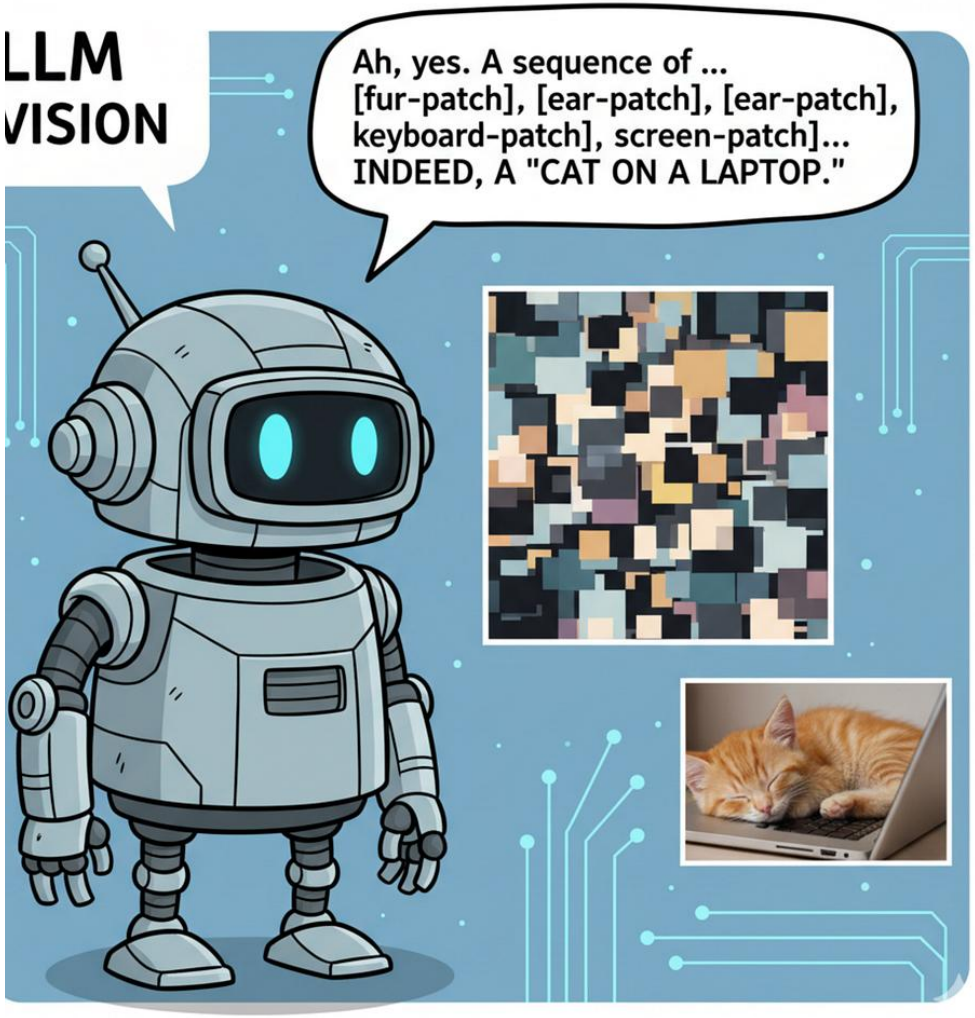
✅ What Text-Only LLMs Do Well

- Open-domain question answering
- Code generation and debugging
- Text summarisation and translation
- Multi-step chain-of-thought reasoning
- Few-shot learning from context examples
- Creative writing and structured dialogue

✗ What Text-Only LLMs Cannot Do

- Process or understand images, audio, video
- Answer questions about visual content
- Read charts, diagrams, or screenshots
- Describe what is in a photograph
- Interpret medical scans or documents
- Perform OCR without external tools

To handle images, we must extend LLMs with vision — the subject of Part 2



Part 2: The Multimodal Shift

How can something that doesn't have eyes, see?

Let’s start with an uncomfortable truth, LLMs can’t process images directly. They were built for words.

An image is just a huge grid of pixel values, and to a language model, that’s gibberish. So, before your photo of a croissant becomes “A flaky pastry on a plate,” it must be translated into something the model can read: **tokens**.

This translation happens through a component called a **Vision Transformer (ViT),** essentially, the model’s eyes.

Computer vision

Computer vision systems break down visual information into pixels and analyse patterns using machine learning models.

The foundational tasks include:

Image Classification: Identifying the primary subject within an image.

Object Detection: Pinpointing the exact location of multiple objects, often by drawing bounding boxes.

Image Segmentation: Classifying every single pixel in an image to understand boundaries and shapes.

Feature Matching: Recognizing specific shapes, edges, or colours to compare different images

Real-World Applications

This technology powers many modern conveniences and industry advancements:

Autonomous Vehicles: Powers navigation, pedestrian detection, and obstacle avoidance by processing real-time video feeds.

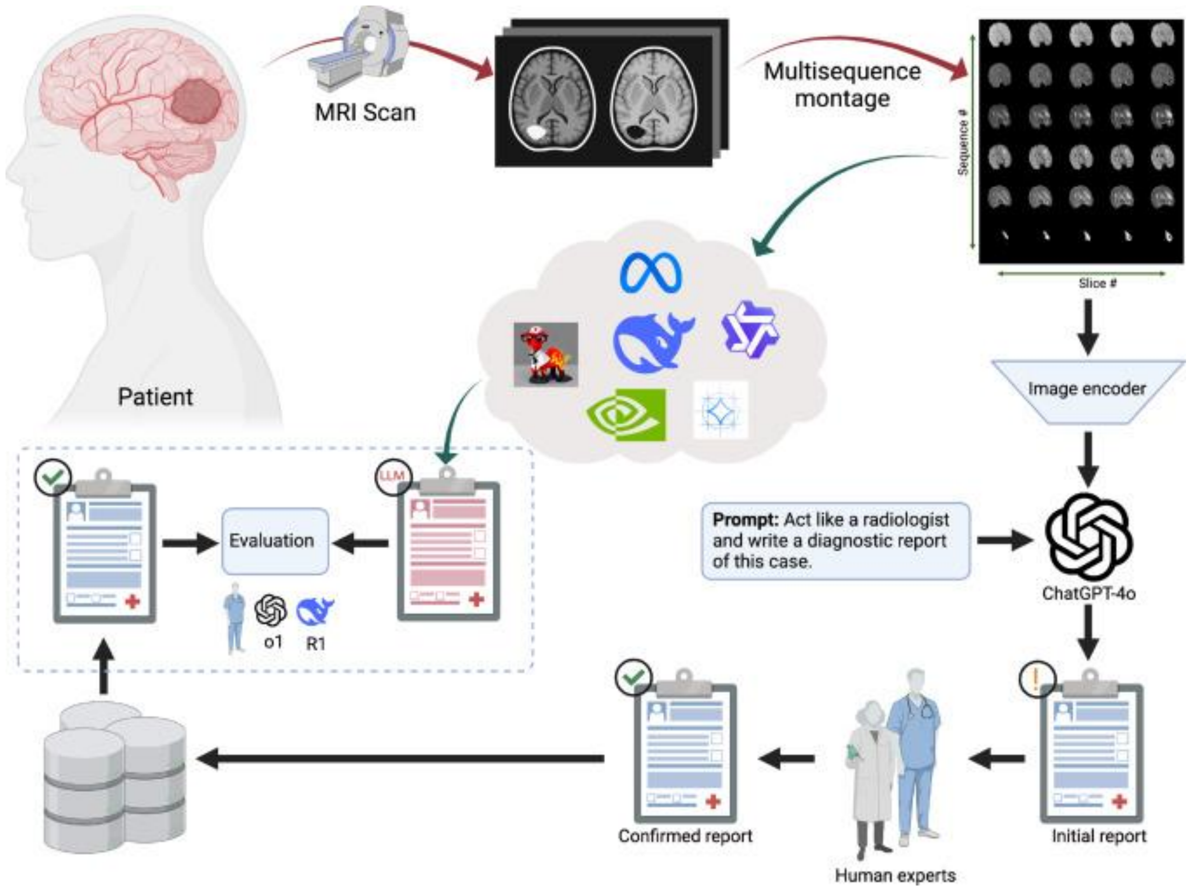
Healthcare: Analyses medical scans (like X-rays or MRIs) to assist radiologists in spotting abnormalities or diseases.

Security & Facial Recognition: Provides automated identity verification and surveillance monitoring.

Manufacturing: Scans assembly lines for product defects and quality control issues at scale.

Real-World Applications

Using VLMs to assist in diagnosing brain tumours



Imaging and Montage: A patient undergoes an MRI scan, which is processed into a multisequence montage to provide comprehensive visual data.

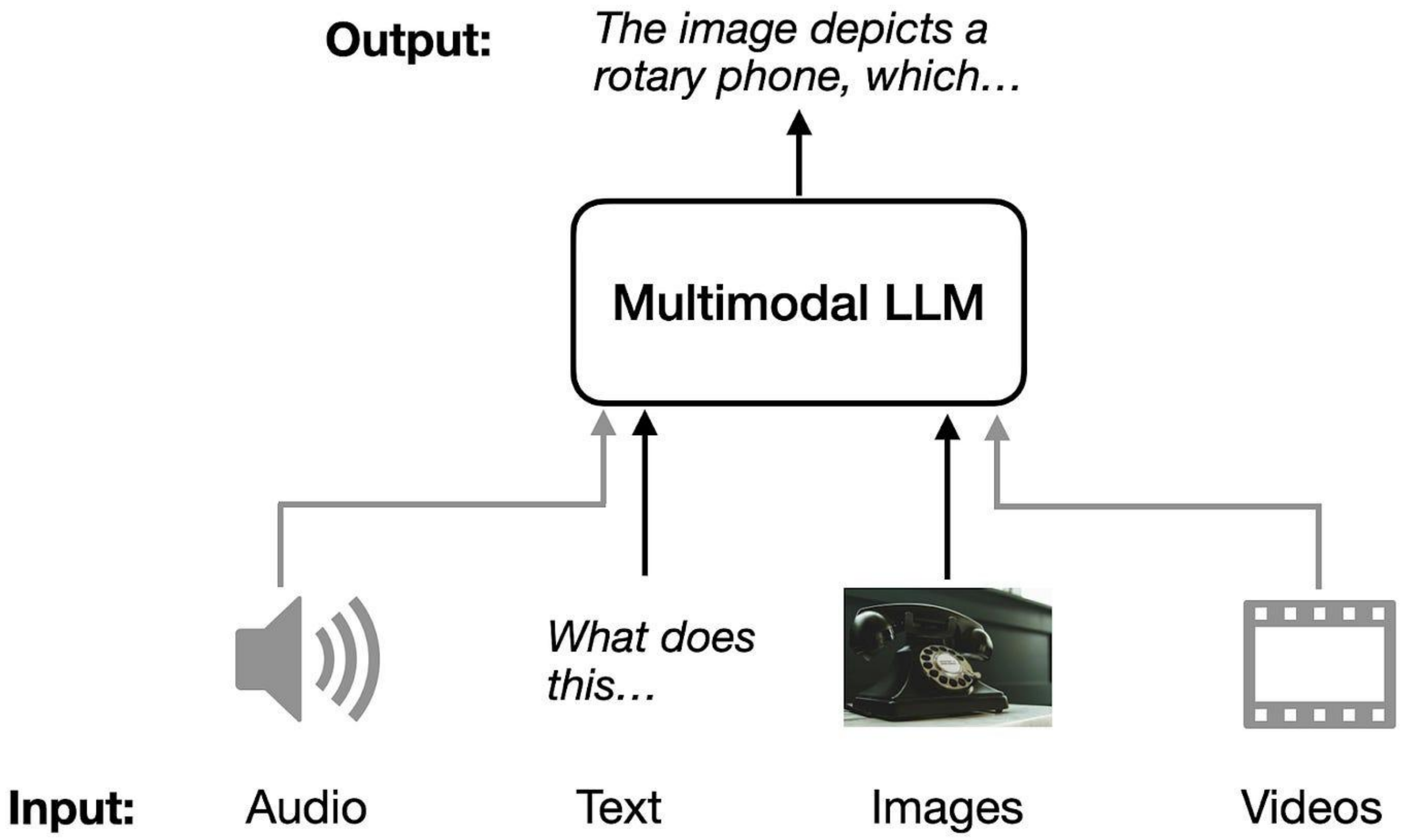
AI Analysis: The montage is analysed by an image encoder and processed by a VLM prompted to act as a radiologist to generate an initial diagnostic report.

Human Review and Confirmation: The initial AI-generated report is reviewed by human experts, who then produce a confirmed diagnostic report.

Evaluation: The system includes a feedback loop for evaluating the reports, potentially involving further review by experts or other AI models.

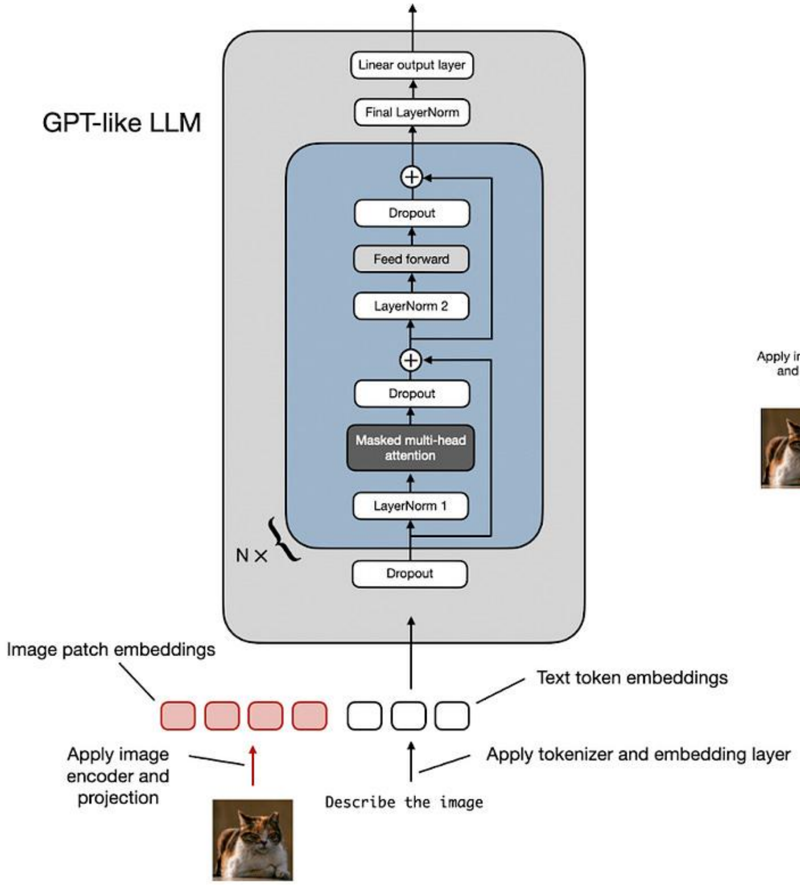
Elboardy AT, Khoriba G, al-Shatouri M, Mousa M, Rashed EA. Benchmarking vision-language models for brain cancer diagnosis using multisequence MRI. Informatics in Medicine Unlocked. 2025 Sep 13:101692.

Multimodal Language Models

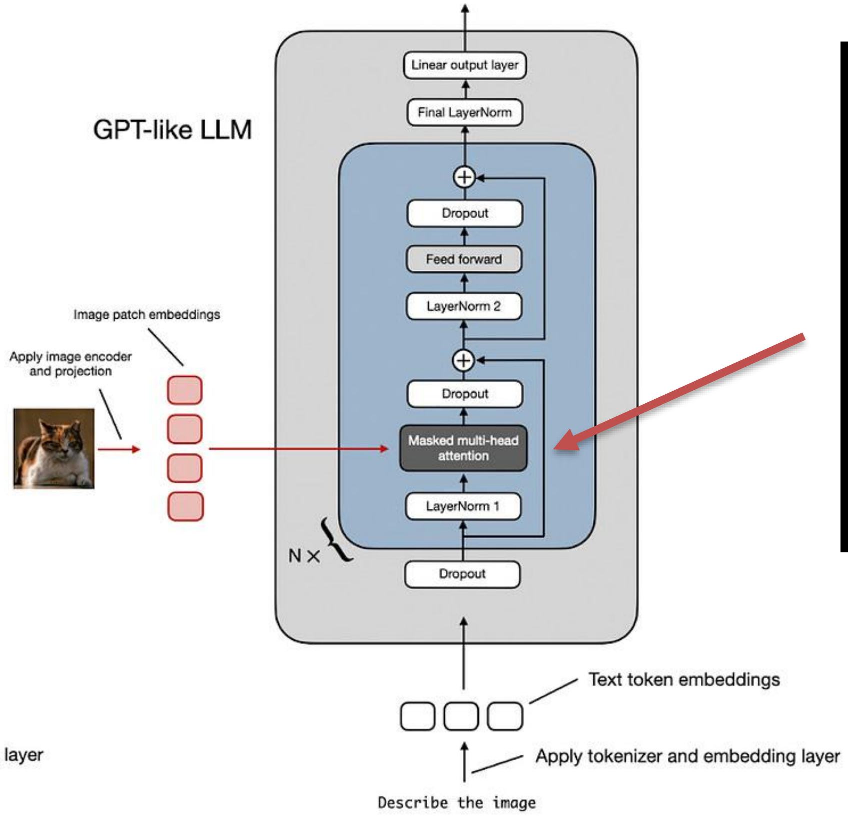


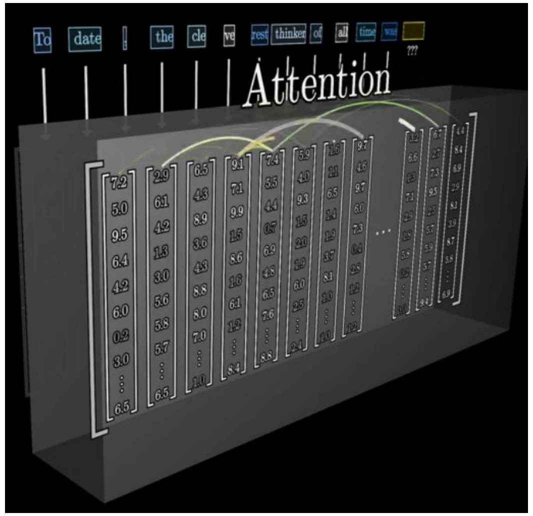
Two main approaches to building Multimodal LLMs

Method A: Unified Embedding Decoder Architecture



Method B: Cross-Modality Attention Architecture





Unified Embedding-Decoder Architecture utilises a single decoder model, much like an unmodified LLM architecture such as GPT-2 or Llama 3.2

Historical Evolution: From CNN to Multimodal Transformers

- CNN Era (2012–2015): AlexNet, ResNet dominate image classification; localised convolution filters learn edge → texture → object hierarchies
- Show & Tell (2015): CNN encodes image to a fixed vector; LSTM decoder generates captions — first end-to-end image captioning
- Limitations of CNN+RNN: sequential LSTM, fixed-length image encoding loses spatial detail, not scalable (it cannot be parallelized)
- ViT (Dosovitskiy 2020): images as sequences of patches — standard transformer, no convolutions — scales like NLP models
- CLIP (Radford 2021): contrastive pre-training of image+text encoders on 400M web pairs — shared embedding space
- Flamingo (2022), BLIP-2 (2023), LLaVA (2023), Llama 3.2 (2024): progressive fusion of ViT-based vision into powerful LLMs

Timeline

- 2012: AlexNet — deep CNN for classification
- 2015: Show & Tell — CNN + LSTM captioning
- 2017: Transformer — attention is all you need
- 2020: ViT — images are patch sequences
- 2021: CLIP — joint vision-language space
- 2022: Flamingo — LLM with cross-attention vision
- 2023: BLIP-2 — Q-Former bottleneck
- 2023: LLaVA — simple projection + SFT
- 2024: Llama 3.2 — frozen LLM + ViT-H/14

Why Images Are Hard for Language Models

- Language models process discrete tokens — images are continuous, high-dimensional pixel arrays
- A 224×224 RGB image contains 150,528 pixel values: no natural token boundary exists
- Naive approach: flatten pixels and embed — results in sequences 100× longer than text, quadratic attention cost
- Semantic gap: raw pixels have no inherent linguistic meaning — 'cat' and a cat photo live in different spaces
- The solution: a Vision Encoder translates visual content into a sequence of patch tokens the LLM can process
- This is exactly what the Vision Transformer (ViT) does — the model's ‘eyes’

The Fundamental Problem

- Text token: discrete symbol, finite vocabulary (~128K)
- Image: continuous real-valued array, infinite variability
- Language model input: fixed-length dense vector per token
- Image patch embedding: map each patch to same vector space
- After patch embedding: image ≈ a sequence of visual tokens
- Same self-attention mechanism can then process vision + text together

Early Fusion: CNN + RNN Show-and-Tell Architectures

- CNN Encoder: deep network (ResNet, Inception) encodes the image into a fixed-length feature vector from the final pooled layer
- LSTM Decoder: the image vector initialises the hidden state; LSTM generates caption tokens one by one, attending to the context
- Problem 1 — Fixed bottleneck: the entire image is compressed to a single vector; spatial detail and object relationships are lost
- Problem 2 — Sequential generation: RNN/LSTM processes tokens one at a time; cannot be parallelised during training
- Problem 3 — Alignment: no mechanism to attend to different image regions when generating different words
- Solution — Attention over CNN features (Xu 2015): soft attention weights over spatial grid; each word attends to relevant regions

Architecture

- Image → CNN → 2048-dim vector
- Vector → LSTM initial hidden state h_0
- Token prediction: y_t = softmax(W_h h_t)
- h_t = LSTM(h_{t-1}, [y_{t-1}; image_vec])
- Spatial attention
- Context
- Limitation: V is flat CNN grid, no global context
- Replaced by: ViT patch tokens + transformer

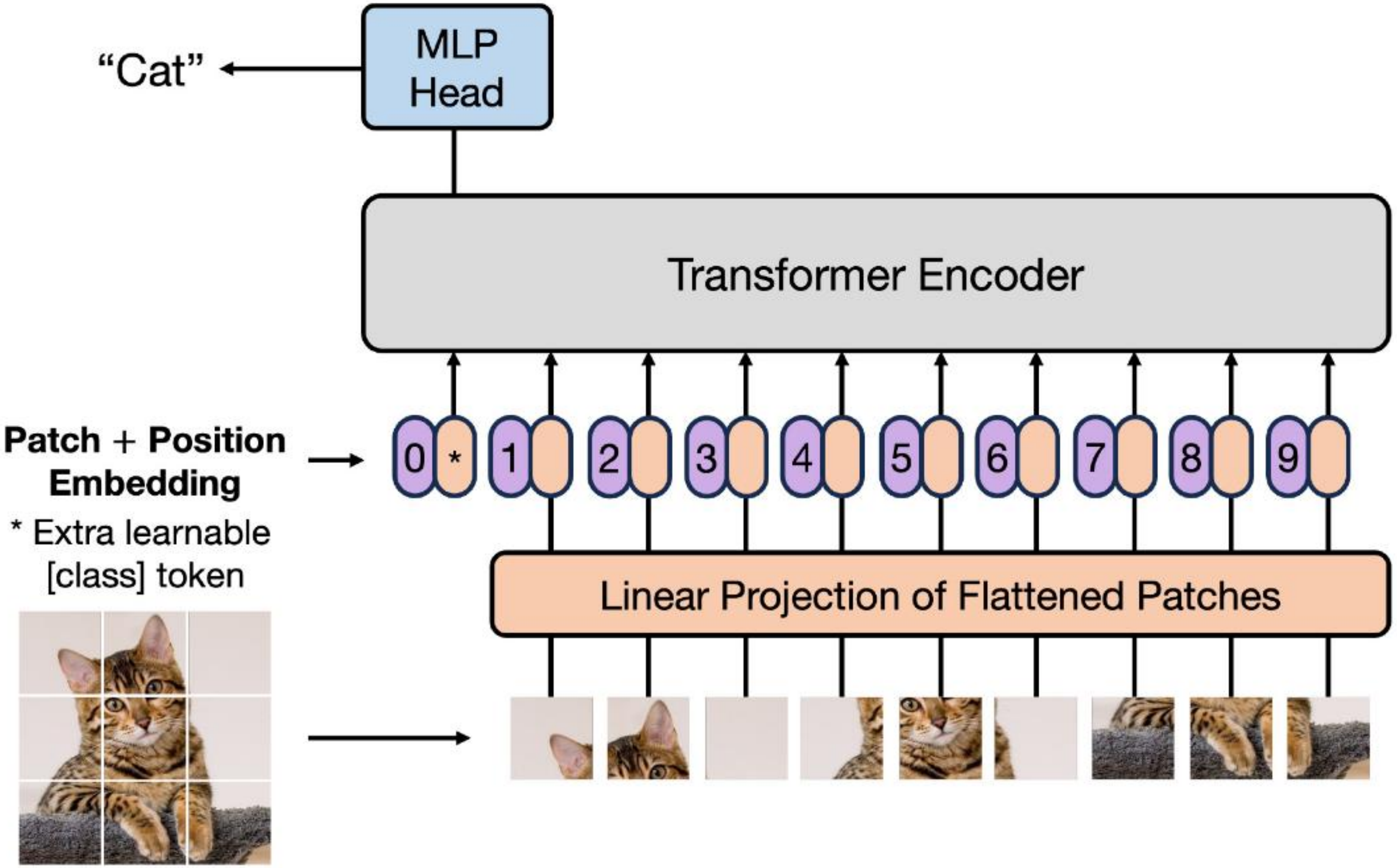
Approaches to Image Tokenisation

- Patch Embeddings (ViT-style): divide image into P×P patches; linearly project each to a D-dimensional vector
- Example: 224×224 image with 16×16 patches → 196 tokens (14×14 grid) — each a 768-dim vector
- Vector Quantisation (VQ-VAE, DALL-E): train a codebook of discrete visual tokens; encode image as token IDs
- VQ approach: enables image generation but requires codebook training; typically 256×16×16 = 1024 codes per image
- Convolutional Features: CNN extracts spatial feature maps; flatten regions into token sequences
- Hybrid: CNN extracts local features, then ViT adds global context — used in some detection architectures

Key Trade-offs

- Patch size ↑ → fewer tokens, lower cost, less detail
- Patch size ↓ → more tokens (8×8: 784 tokens), quadratic cost
- VQ: compact (integer IDs), generation-ready, needs codebook
- Continuous ViT embeddings: richer, better for understanding
- Llama 3.2 uses 14×14 patches → 1,600 tokens per 560×560 tile
- Context window cost: 5 tiles × 1,600 = 8,000 vision tokens

Approaches to Image Tokenisation



Dosovitskiy A, Beyer L, Kolesnikov A, Weissenborn D, Zhai X, Unterthiner T, Dehghani M, Minderer M, Heigold G, Gelly S, Uszkoreit J. An image is worth 16x16 words: Transformers for image recognition at scale. arXiv preprint arXiv:2010.11929. 2020 Oct 22.

Patch Embeddings in Detail: From Pixels to Tokens

- Step 1 — Slice: divide $H \times W \times C$ image into $(H/P) \times (W/P)$ non-overlapping patches of size $P \times P \times C$ each
- Step 2 — Flatten: reshape each patch from $P \times P \times C$ to a 1D vector of length $P^2 \times C$ (e.g., $16 \times 16 \times 3 = 768$ values)
- Step 3 — Project: multiply by learned weight matrix $W_e \in \mathbb{R}^{(P^2 C \times D)}$ → D-dimensional embedding
- Step 4 — Positional encoding: add 2D sinusoidal or learned positional embeddings to preserve spatial layout
- Step 5 — CLS token (optional): prepend a learnable [CLS] token; its final hidden state represents the whole image
- Result: sequence of $N = (H \times W)/P^2$ tokens fed to transformer self-attention

Vision Transformer (ViT): Images as Sequences

- Core insight (Dosovitskiy 2020): bypass spatial convolutions entirely — treat 2D image as a 1D sequence of patches
- Sequence framing: N patch embeddings flow through standard transformer encoder blocks — no architectural changes
- Global self-attention: each patch attends to every other patch — captures long-range spatial dependencies (e.g., object parts across image)
- Unified foundation: same transformer backbone processes both text tokens and image patch tokens
- Scaling: ViT-B (86M params, 12 layers, 768 dim) → ViT-L (307M) → ViT-H (632M, 32 layers, 1280 dim)
- Pre-training: originally supervised on ImageNet/JFT; modern VLMs use CLIP-style contrastive pre-training

ViT vs CNN

- CNN: inductive bias (locality, translation equivariance)
- ViT: no inductive bias — learns from data
- CNN: good on small datasets, faster convergence
- ViT: better at scale, requires large pre-training data
- ViT-H/14 (Llama 3.2): 14×14 patches, 32 layers, 16 heads
- Patch size 14: fine-grained — captures text, medical details
- Training: 6B image-text pairs — rivals or exceeds CNN

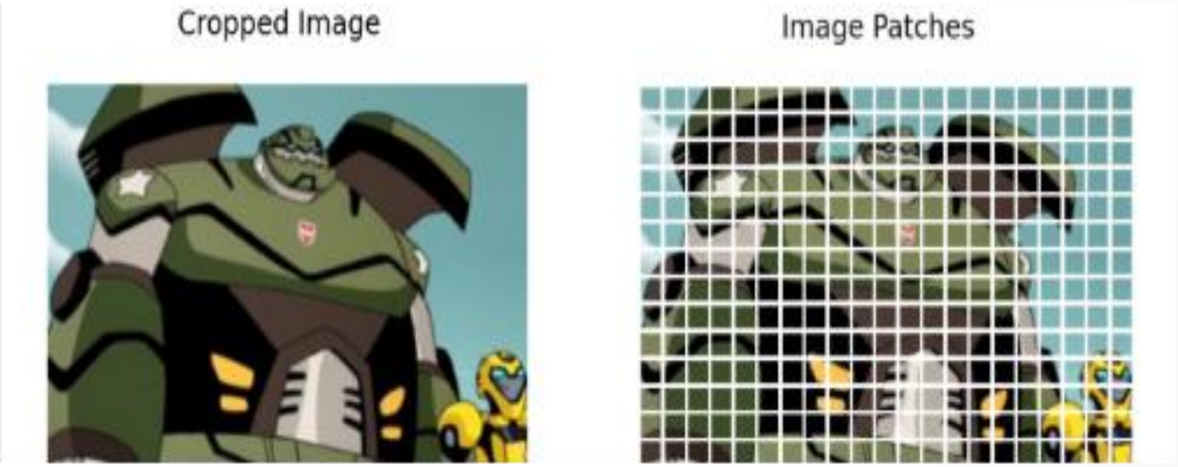
Vision Transformer (ViT)

ViT completely bypassed spatial convolutions, reframing 2D visual frames as standard sequence modeling pipelines.

Sequence Framing: Treats distinct image patches as if they were sequential language words.

Global Self-Attention: Replaces local constraints with global processing.

Unified Foundation: Aligns representation structures, allowing identical transformer blocks to process both images and texts.



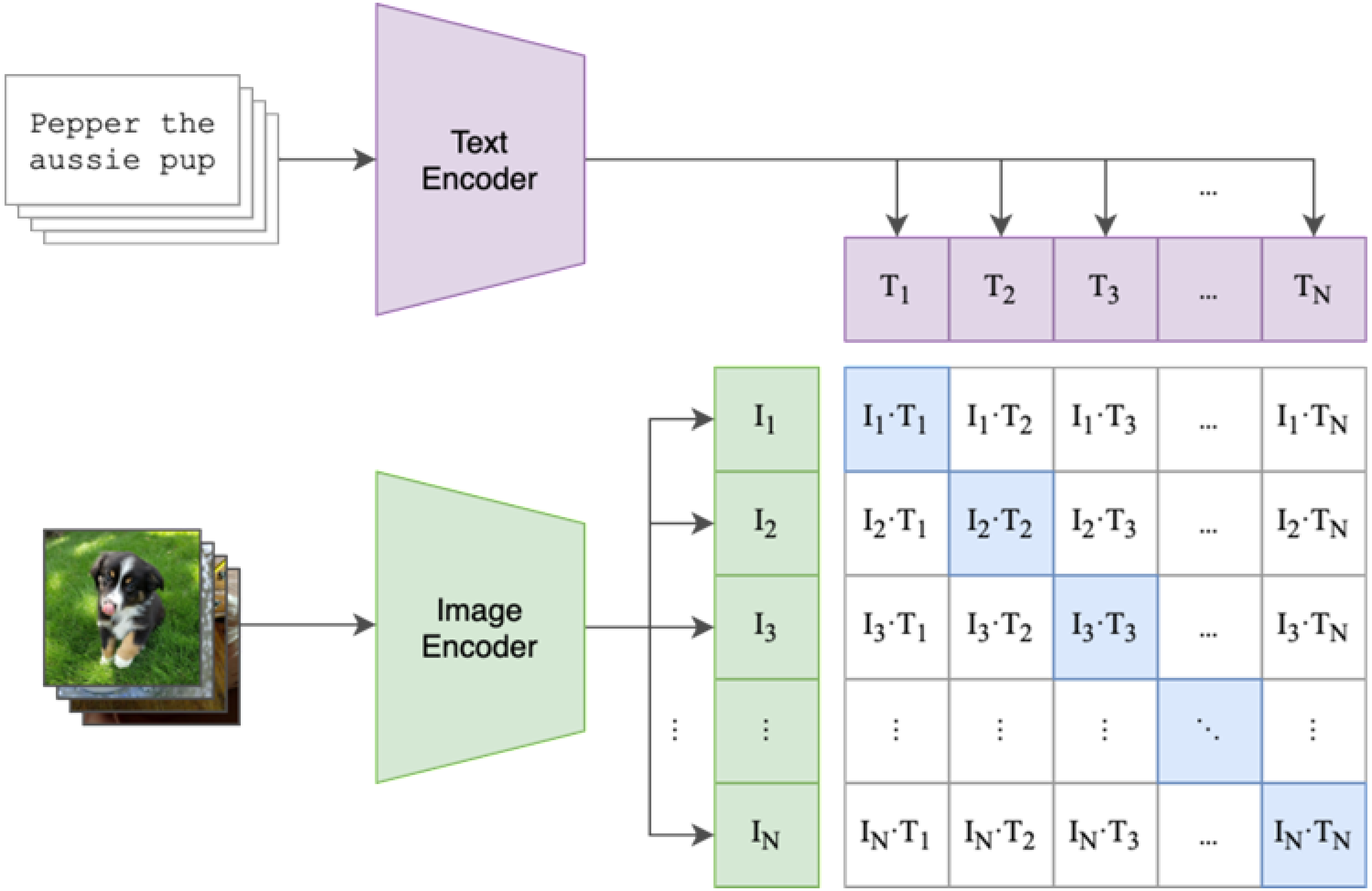
CLIP: Contrastive Language–Image Pre-Training

- Proposed by Radford et al. (OpenAI, 2021) — trained on 400M **image-text pairs** from the internet
- Architecture: separate image encoder (ViT or ResNet) and text encoder (Transformer); both output L2-normalised vectors
- Training objective: contrastive — maximise similarity of matched (image, text) pairs
- Loss: InfoNCE (noise-contrastive estimation) over N×N similarity matrix; temperature-scaled softmax per row and column
- Result: a shared embedding space where semantically related image and text vectors cluster together
- Zero-shot classification: encode class names as text prompts; compare image embedding to each class; pick nearest

Why CLIP Matters

- No task-specific labels needed — uses existing image-text pairs
- Versatile: same model for classification, retrieval, generation guidance
- Foundation: almost every VLM built since 2021 uses CLIP or a CLIP-derived encoder
- Medical: BioViL-T, PubMedCLIP trained on radiology image-report pairs
- Limitation: weak on fine-grained counting, spatial relations, novel compositions

CLIP: Contrastive Language–Image Pre-Training



[OpenAI]

The CLIP Embedding Space: Language-Aligned Vision

- CLIP creates a shared embedding space where text and image representations are geometrically aligned
- Semantic richness: CLIP features capture complex visual-linguistic relationships, not just object categories
- Shared projections: image vectors occupy coordinates that map naturally to linguistic concepts in text space
- Generalisation: trained on 400M diverse image-text pairs — broad coverage of visual concepts in natural language
- Downstream transfer: CLIP image features serve as a powerful initialisation for multimodal VLMs
- Nearly all modern VLMs (Flamingo, BLIP-2, LLaVA, Llama 3.2) use CLIP or CLIP-derived encoders as their vision backbone

CLIP in VLMs

- Direct use: extract image features, pass to LLM
- Fine-tuned CLIP: domain adaptation (medical, satellite)
- BioViL: CLIP trained on chest X-ray / radiology reports
- PubMedCLIP: fine-tuned on biomedical literature figures
- Contrastive + generative: CLIP + GPT for unified generation
- Key property: language-aligned — image embeddings live near the text descriptions of what they show

Why Language-Aligned Features Are Critical

- Without alignment: image features (CNN activations) and text embeddings live in completely separate vector spaces
- With alignment (CLIP): 'a chest X-ray showing pneumonia' and the actual X-ray image map to nearby vectors
- Plug-and-play: CLIP features can be directly projected into any LLM's embedding space with minimal training
- Abstraction level: CLIP captures semantic content (objects, scenes, relationships) — not raw pixel statistics
- The CLIP projection adapter (linear layer or MLP) maps from vision to language dimension: ℝ^{1280} → ℝ^{4096}
- After projection: the LLM sees visual tokens that behave like semantically meaningful text tokens — enabling genuine vision-language reasoning

Alignment Quality

- Dog photo ↔ 'a photo of a dog': cosine sim ≈ 0.28
- Unrelated images: cosine sim ≈ 0.0–0.05
- Medical: BioViL shows abnormal CXR ↔ 'right lower lobe consolidation' align well after domain FT
- Retrieval: given text query, find closest image in embedding space
- Robustness: text prompts like 'an X-ray of...' outperform 'X-ray:' prefix
- Failure: counting, spatial prepositions, fine-grained part distinction

03

Image Description with LLMs

Captioning · VQA · Chart Understanding · Spatial Reasoning · Failure Modes

Model Collection Overview

1, 3, 11, 90 B

Text only

Drop-In Vision Capabilities

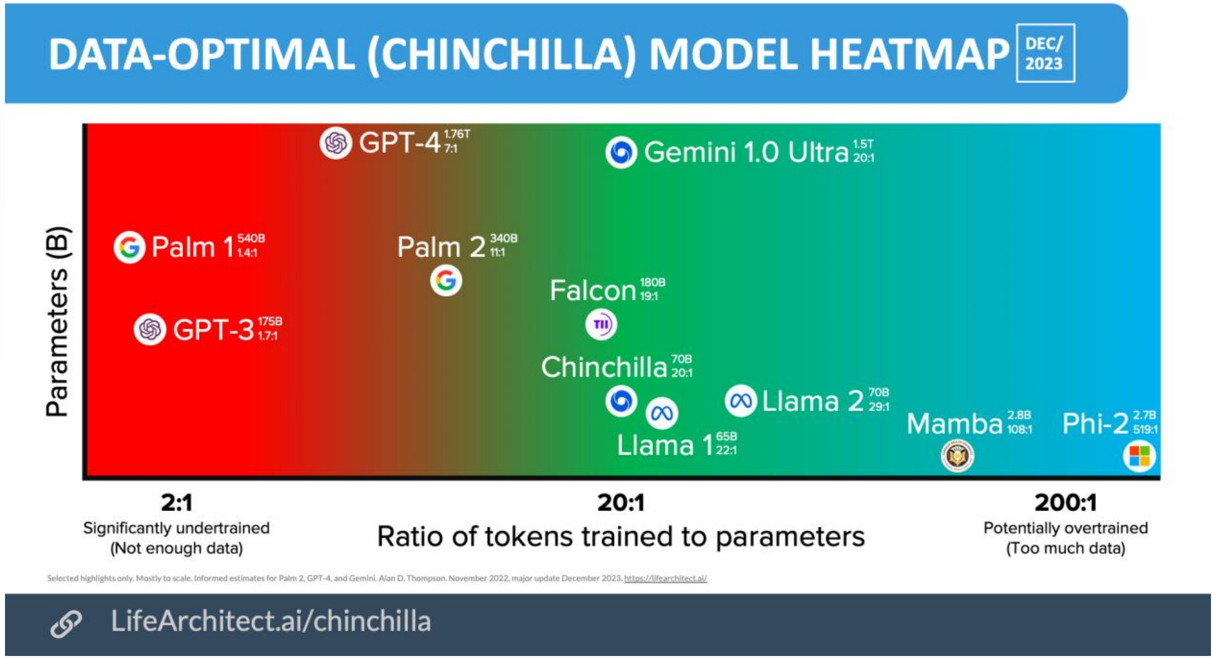
Meta's Llama 3.2 family introduces high-performance 11B and 90B multimodal models engineered for visual recognition, document QA, and reasoning.

Equipped with a native 128K context window and Grouped-Query Attention (GQA), these structures serve as complete, drop-in multimodal replacements for the text-only Llama 3.1 counterparts.

Key Metrics & Scaling

15T

Pre-training Tokens



The Power of Chinchilla Scaling

Modern LLMs require a balanced scaling law to achieve optimal parameter performance. Training compute scales jointly with dataset size and total model parameters.

Llama 3 base structures leverage up to 15 trillion highly filtered tokens, ensuring that even relatively small parameter models (like 8B) reach an exceptionally high asymptote of generalized world knowledge before visual adapters are integrated.

Cross-Attention Design

Why Meta Avoided Concatenation

Concatenating image tokens into the self-attention sequence directly disrupts established positional representations, risking degradation of the text model's core math and reasoning performance.

Additionally, image tokens consume substantial context window space, limiting document length and increasing compute overhead.

The Llama 3.2 Integration Strategy

By using dedicated cross-attention blocks, the original language weights stay completely unchanged. During inference, visual features are processed once and stored in a distinct visual memory cache.

This allows text generation to run token-by-token with zero language degradation, preserving the full text context window.

Example 1: Image Captioning

Task: Generate a natural language description of an image

Prompt: 'Describe this image in detail.'

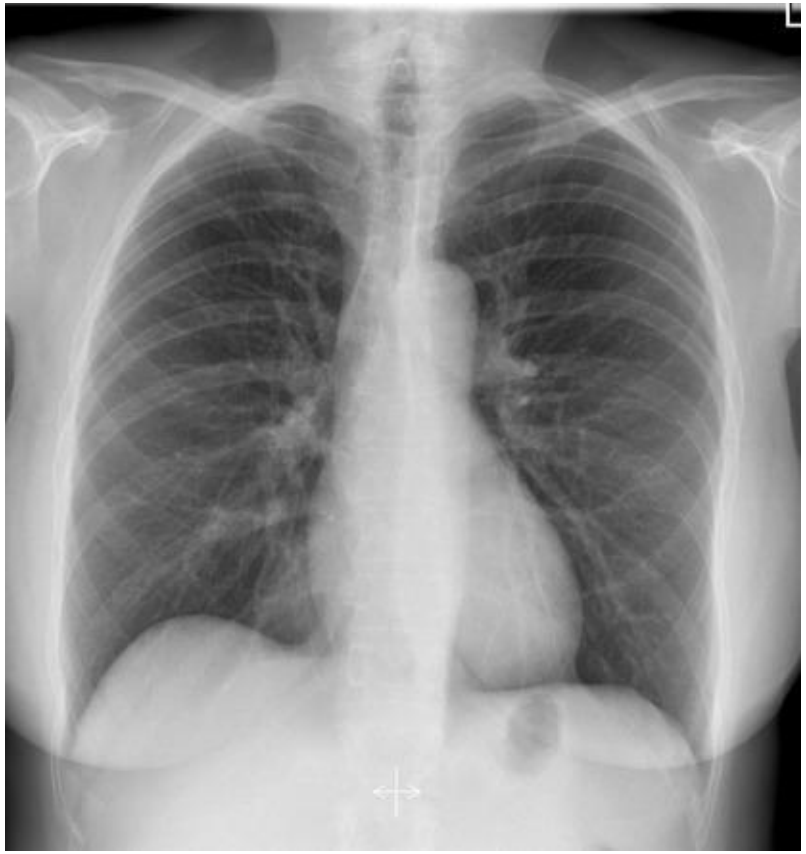
✅ Fine-Tuned Model Output:

Input: photograph of a busy city street at night — neon signs, taxi cabs, pedestrians crossing

Output: 'A vibrant night-time urban scene shows a busy intersection bathed in the glow of neon signs in red, yellow, and blue. Several yellow taxi cabs wait at a traffic light while crowds of pedestrians cross in both directions. Tall glass office buildings reflect the city lights, and a street vendor cart is visible in the lower left.'

— Fluent, grounded, spatially aware description

Example 1: Image Captioning



Medical Captioning (after fine-tuning)

Prompt: 'Describe this PA chest radiograph.'

✅ **Model Output:**

Before FT: 'Black and white torso image.’

After FT: 'PA chest radiograph. Lung fields clear bilaterally. Cardiac silhouette normal. No pleural effusion. No consolidation.
Costophrenic angles sharp.’

→ Domain fine-tuning dramatically improves clinical specificity and vocabulary.

Captioning is the foundation: if a model can describe an image accurately, it can answer questions, extract information, and reason about content. Quality degrades for unusual compositions, low-resolution inputs, and domain-shifted images (e.g., medical scans, satellite imagery, historical documents).

Example 2: Visual Question Answering (VQA)

🖼 Image: Bar chart — monthly sales for Products A, B, C (Jan–Jun)

Q: Which product had highest sales in March?

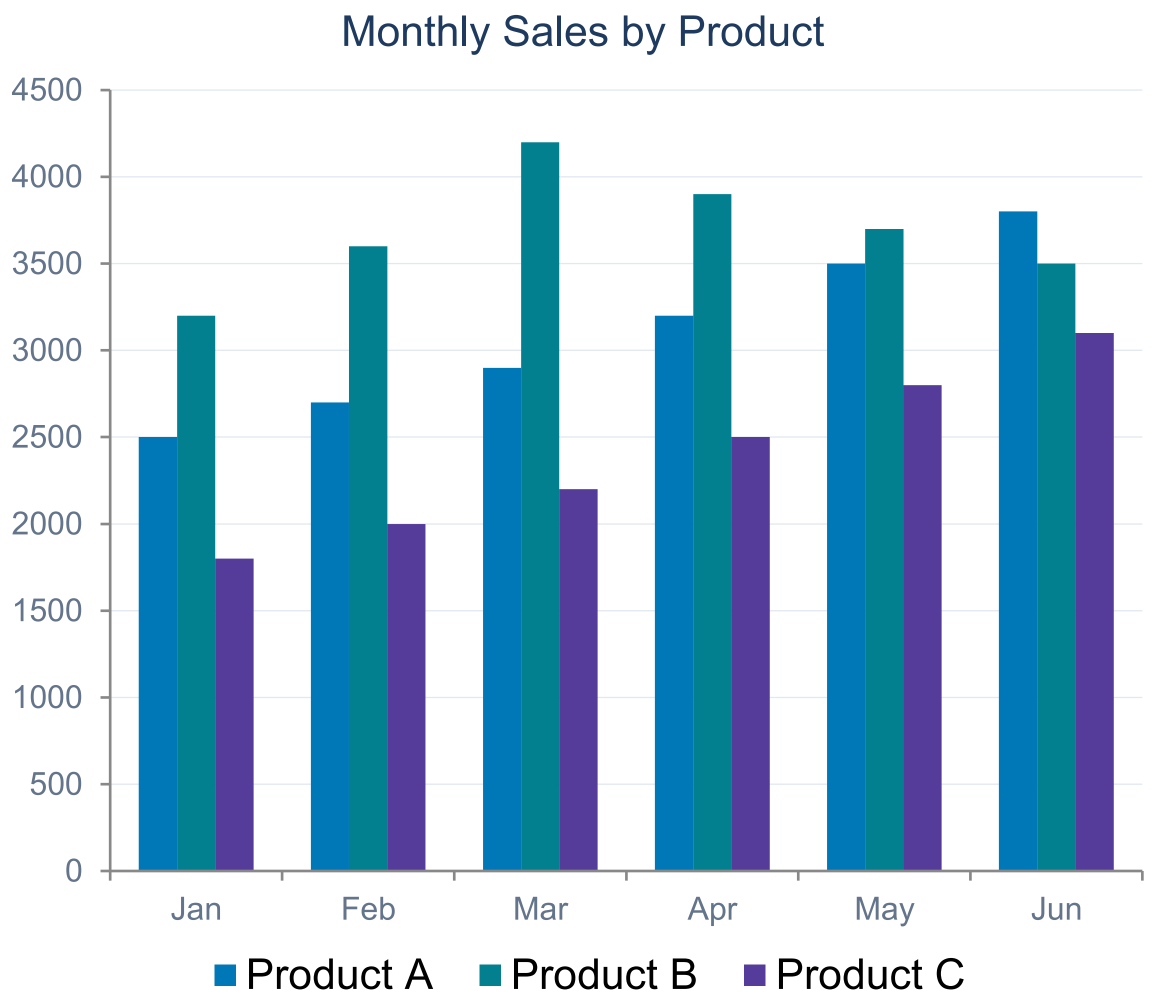
A: Product B peaked at ~4,200 units in March, the highest of all three products that month.

Q: What is the overall trend for Product C?

A: Product C shows a steady upward trend, growing from ~1,800 units in January to ~3,100 in June (+72%).

Q: How much did Product A grow from Jan to Jun?

A: Product A grew from ~2,500 to ~3,800 units — an increase of ~1,300 units (~52% growth).



Example 3: OCR and Document Understanding

- Read text from images, screenshots, photographs, handwriting
- Understand document layout: headers, columns, tables, forms
- Extract and interpret key information from invoices, reports, slides
- Go beyond OCR: understand the meaning of text in context

🖼 *Image: Invoice document with a handwritten annotation*

```
INVOICE #4471   Date: 12/03/2024
Item: Consulting Services —— £2,400.00
Handwritten note: "Total: two thousand four hundred
pounds"
```

Prompt: "Extract all financial info and flag any discrepancies."

✅ **Model Output**

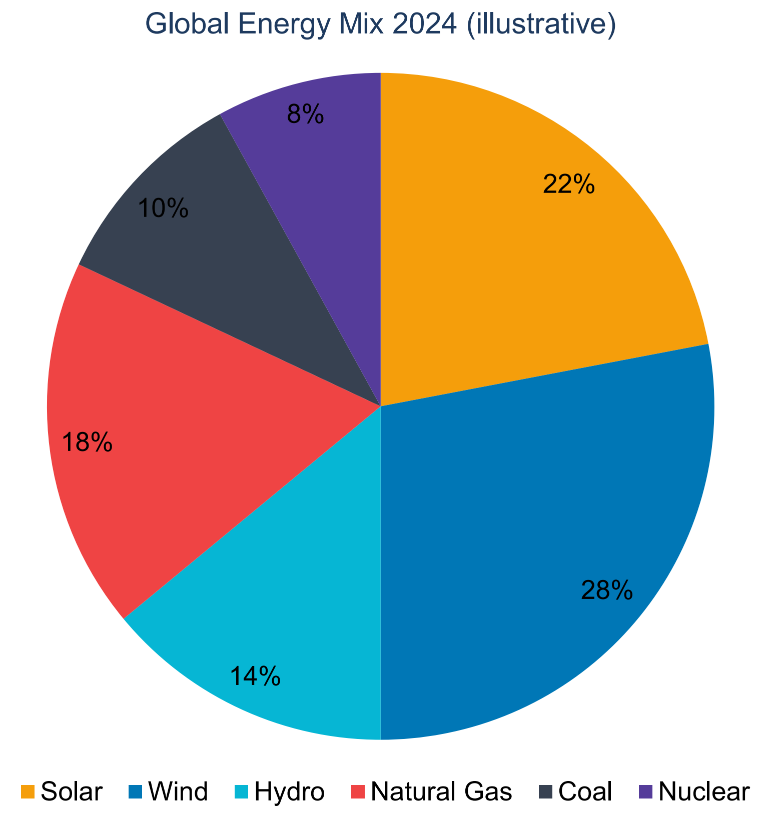
"Invoice #4471, dated 12 March 2024. Item: Consulting Services, £2,400.00. Handwritten annotation confirms total as two thousand four hundred pounds — consistent with the printed amount. No discrepancy detected."

Prompt: "What does the handwritten note say?"

✅ "The handwritten note reads: Total: two thousand four hundred pounds."

✗ Failure: cursive handwriting or low-resolution scans increase error rate. Verify critical financial/legal documents.

Example 4: Chart and Diagram Understanding



Prompt: "Summarise the key findings from this energy chart."

✅ **Model Output (Correct)**

"The chart presents a hypothetical 2024 global energy mix with several notable patterns:

• Renewables dominate: Wind (28%) and Solar (22%) together account for 50% of generation — a historic milestone.

• Fossil fuels contribute 28%: Natural Gas leads at 18%; Coal has the smallest fossil share at 10%.

• Nuclear (8%) and Hydro (14%) provide stable low-carbon baseload.

Key insight: Renewables combined (64%) substantially outpace fossil fuels (28%). Wind alone is the single largest source, surpassing Natural Gas."

Example 5: Spatial Reasoning and Object Counting

🖼 Floor plan: 3 rows × 4 desks (12 total); conference room top-right; 2 fire exits east wall; coffee area bottom-left

Q: How many desks are there in total?

A: 12 desks — 3 rows of 4 columns each. ✅

Q: Is the conference room adjacent to an exit?

A: The conference room is in the top-right corner; exits are on the east (right) wall — they are on the same side but not directly adjacent. ✅

Q: Which desks are furthest from the coffee area?

A: The top-right corner desks near the conference room are the farthest from the bottom-left coffee area. ✅

✗ Known Failure: Counting in Dense Scenes

"Image: Dense crowd of ~213 people"

Prompt: How many people are in the image?

Model: "Approximately 150–180 people."

✗ True count: 213. LLMs systematically under-count in crowded or occluded scenes.

⚠ Additional Spatial Limitations

- Exact pixel coordinates are unreliable
- Left/right confusion in mirrored images
- Depth estimation is approximate only
- Exact angular or metric measurements not possible

Failure Modes: Hallucination in Multimodal LLMs

Hallucination: the model generates plausible-sounding but visually incorrect content

Object Hallucination

"Image: A cat on a blue sofa"
Model: "A dog and a cat sit on a red sofa."
(no dog; colour is blue, not red)

Attribute Error

"Image: A green apple on a table"
Model: "The ripe red apple appears fresh."
(apple is clearly green)

Text Misread (OCR error)

"Image: Road sign reading STOP"
Model: "The sign reads SHOP."
(single character recognition error)

Spatial / Relational Error

"Image: Cat sitting to the LEFT of a box"
Model: "The cat is sitting on top of the box."
(wrong spatial relationship)

Mitigation: prompt for explicit evidence, apply output verification, use hallucination detection pipelines in production.

Evaluating Multimodal LLMs: Benchmarks and Metrics

Standard Metrics

BLEU / ROUGE

N-gram overlap with reference captions — fast but surface-level

CIDEr

Consensus-based similarity; rewards human-like descriptions

CLIPScore

Cosine similarity between image embedding and generated text

VQA Accuracy

Exact match on closed-set visual questions

Human Evaluation

Expert rater scores: accuracy, completeness, actionability

Key Benchmarks

Benchmark	What It Tests
VQAv2	General visual question answering
TextVQA	Reading text in natural images (OCR)
MMMU	Multi-discipline university-level VQA
DocVQA	Document understanding
ChartQA	Chart comprehension and reasoning
MMBench	Comprehensive multi-task evaluation
HallusionBench	Hallucination detection rate

Medical Image Description: Why It Changes Clinical Practice

- Clinical relevance: a model that can accurately describe a chest X-ray, CT slice, or histology slide provides immediate diagnostic support
- Speed: a radiologist reads 20–30 complex studies per day; AI can draft structured reports in seconds — reducing reporting backlog
- Completeness: studies show AI catches findings missed in busy reading sessions — second-reader effect
- Standardisation: AI outputs follow consistent report structure (Findings, Impression, Differential) — reduces inter-reader variability
- Triage: flagging urgent findings (pneumothorax, intracranial haemorrhage) before the radiologist opens the case
- Education: trainees can compare their impression to the model's structured output — accelerates learning

The Clinical Gap

- General VLMs on chest X-rays: 'shows some dark and light areas'
- Clinically fine-tuned VLMs: structured Findings + Impression
- Key challenge: medical vocabulary, normal anatomy knowledge, abnormality patterns
- Dataset bottleneck: high-quality annotated medical image-text pairs are scarce and expensive
- Regulatory barrier: clinical deployment requires FDA/CE approval, validation studies
- Opportunity: Part 5 of this lecture shows how to bridge this gap with fine-tuning

04

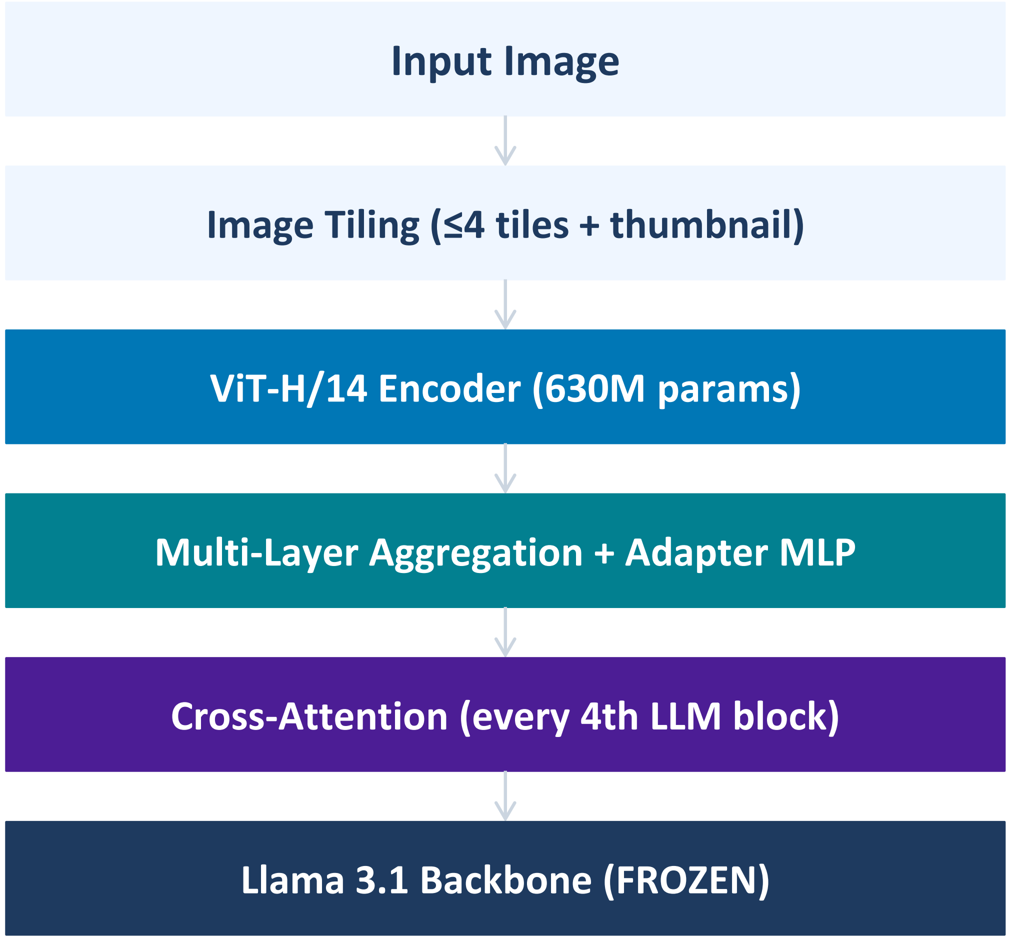
Llama Vision Deep Dive

Image Tiling · Cross-Attention · 4-Stage Training Pipeline

Llama 3.2 Vision: Architecture Overview

- Released by Meta, September 2024 — first open-weight multimodal Llama model
- Two sizes: 11B (Llama 3.1 8B backbone + 3B vision) and 90B (70B + 20B vision)
- Follows Flamingo-style cross-attention — NOT LLaVA concatenation
- Llama 3.1 backbone weights completely frozen throughout all training
- ViT-H/14 vision encoder (630M params) trained from scratch on 6B image-text pairs
- Supports: captioning, VQA, OCR, chart understanding, document parsing (single image)

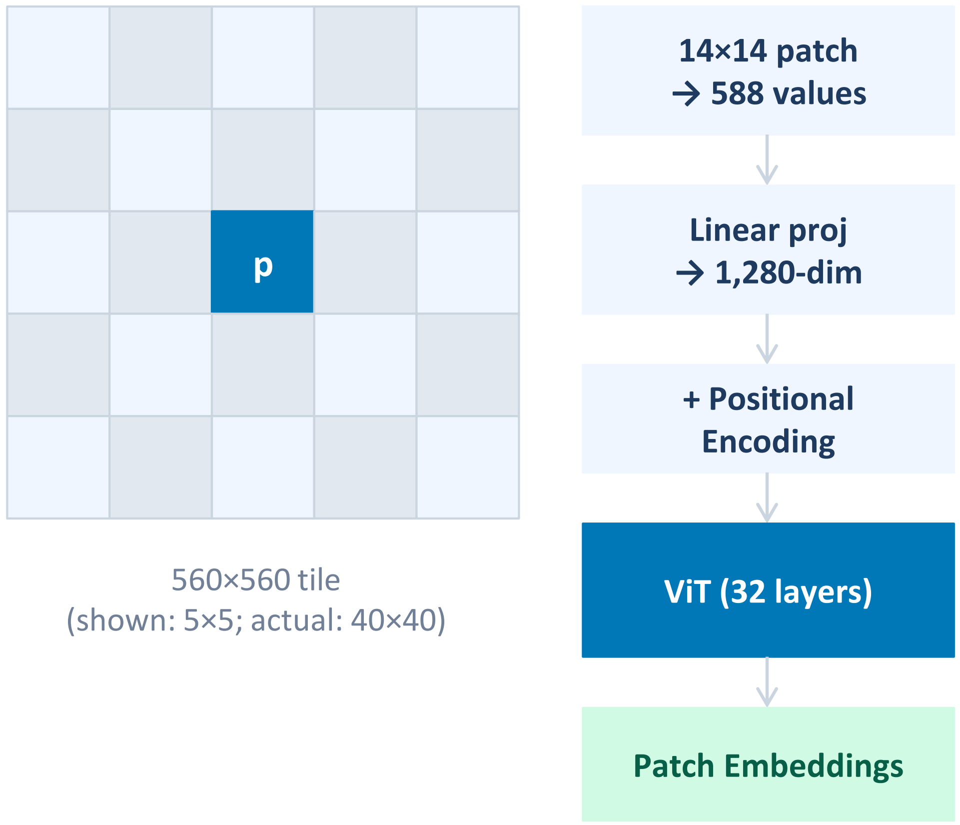
Key advantage: because LLM weights are frozen, Llama 3.2 Vision's text-only performance is bit-identical to Llama 3.1 — vision is added without any text regression.



The Vision Encoder: ViT-H/14 (630M Parameters)

Property	Value
Variant	ViT-H ("Huge")
Patch size	14 × 14 pixels
Input tile size	560 × 560 pixels
Patches per tile	560 ÷ 14 = 40 → 40 × 40 = 1,600 tokens
Hidden dimension	1,280
Attention heads	16
Transformer layers	32
Parameters	~630 million
Pre-trained on	~6 billion image-text pairs (Meta, from scratch)

Patch Embedding Process

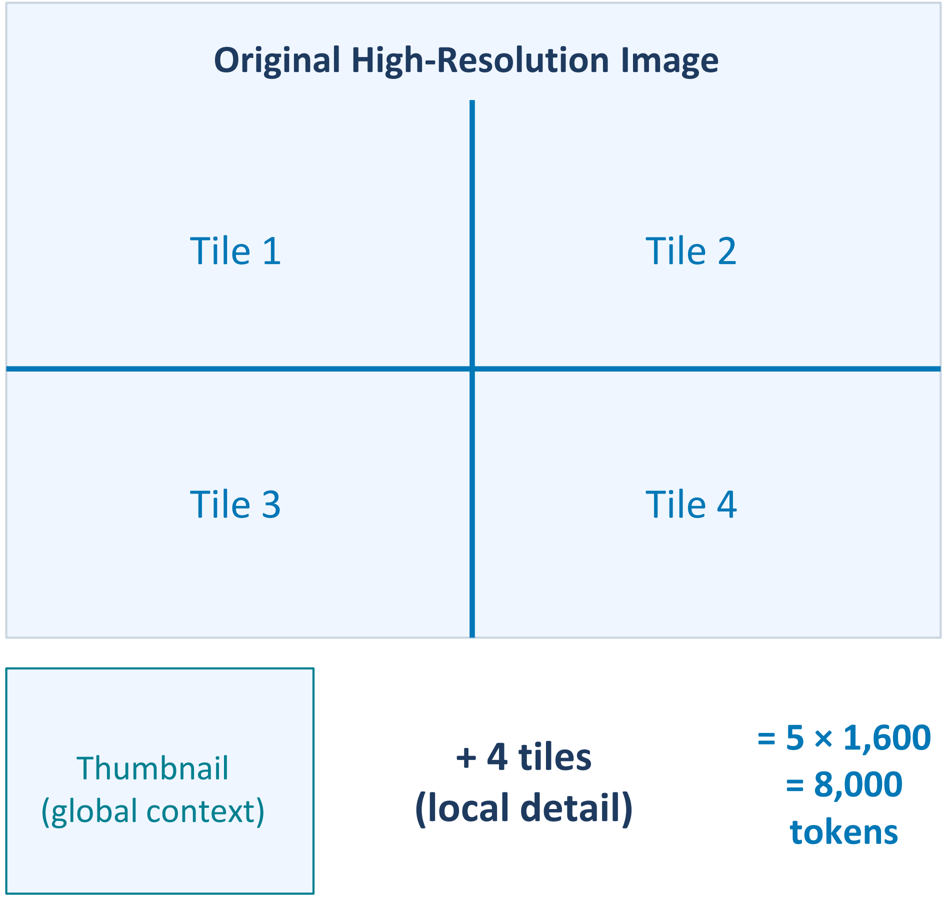


Multi-layer feature aggregation: features from multiple ViT layers are concatenated, capturing both spatial detail (early layers) and semantic content (later layers).

Image Tiling: Handling High-Resolution Input

- Problem: a single 560×560 downsampled tile loses fine detail in large images
- Solution: tile the image into up to 4 tiles (each 560×560) + 1 thumbnail
- Thumbnail provides global context; tiles provide local detail
- 4 tiles + 1 thumbnail = 5 × 1,600 = 8,000 patch tokens fed to ViT-H/14
- Two sets of positional encodings: intra-tile position AND tile-level position
- This enables reading fine text, dense charts, and A4-format documents

Without tiling: a prescriptions label or pathology slide would be unreadable. With tiling, each tile is processed at full 560×560 resolution.



4 tiles (560×560 each) + 1 thumbnail → 8,000 ViT patch tokens

Cross-Attention: The Fusion Mechanism in Llama 3.2

- New cross-attention blocks inserted at every 4th Llama 3.1 decoder layer
- Queries (Q) = linear projection of text token hidden state
- Keys (K) and Values (V) = linear projections of image features (pre-computed once)
- Image features stored as fixed memory — NOT mixed into the text sequence
- Result added to text residual stream (same pattern as self-attention)
- The context window is fully available for text — no image token overhead

This is the key difference from LLaVA: image tokens do NOT consume the context window. A 4,096-token context window remains 100% available for the text prompt.

Text Hidden State → Q

Cross-Attention Block

$Q = W\_q \cdot \text{text\_hidden}$

$K = W\_k \cdot \text{image\_features}$

$V = W\_v \cdot \text{image\_features}$

$\text{Out} = \text{softmax}(QK^T/\sqrt{d}) \cdot V$

Attended Output

Add & Norm → Residual Stream

Image Features (K, V — pre-computed once)

Forward Pass: From Image + Prompt to Output Text

1 **Image Tiling**

Input image tiled into ≤4 tiles (560×560 each) plus a thumbnail

2 **Patch Tokenisation**

Each tile → 40×40 = 1,600 patch tokens via ViT-H/14 encoder

3 **ViT Encoding**

All tiles processed through ViT-H/14 → sequence of image feature vectors

4 **Multi-Layer Aggregation**

Features from multiple ViT layers concatenated; projected by adapter MLP to LLM dimension

5 **Text Tokenisation**

Prompt tokenised normally; image features stored separately as fixed memory

6 **LLM Decoding**

Text tokens flow through Llama 3.1; every 4th block fires cross-attention over image features

7 **Autoregressive Generation**

Model generates output token-by-token; image features reused at each step — computed only once

4-Stage Training Pipeline

Stage 1	**Vision Encoder Pre-Training** ViT trained from scratch on 6B image-text pairs CLIP-style contrastive + auxiliary losses Goal: language-aligned image representations	**Trains: ViT encoder only**
Stage 2	**Cross-Attention Adapter Pre-Training** Llama 3.1 weights frozen Cross-attention layers initialised (gate=0) Large image-text corpus: captions + visual QA Goal: connect vision to LLM	**Trains: cross-attn layers + adapter MLP**
Stage 3	**Supervised Fine-Tuning (SFT)** Curated instruction data: VQA, OCR, charts, documents, captioning, multi-turn image conversations Partial updates to cross-attention components	**Trains: cross-attn (selective updates)**
Stage 4	**Preference Optimisation (DPO)** Rejection sampling + Direct Preference Optimisation Human preference data on vision tasks Aligns for helpfulness, accuracy and safety	**Trains: cross-attn fine-tuning**

Throughout all 4 stages, the original Llama 3.1 self-attention and FFN weights remain completely frozen.

Llama 3.2 Vision: Performance and Hardware

Benchmark	Llama 3.2 11B	Llama 3.2 90B	GPT-4V
MMMU (val)	50.7	60.3	63.1
DocVQA	88.4	90.1	88.4 ← 11B m
ChartQA	83.4	85.5	78.5 ← both e
TextVQA (OCR)	73.8	78.6	78.0
VQAv2	75.2	78.1	77.2 ← 90B le
MMLU (text-only)*	73.0	86.0	(frozen = unch

** Text-only performance preserved from Llama 3.1 thanks to frozen backbone weights.*

Hardware Requirements

11B (4-bit quantised):

→ Single RTX 3090/4090 (24 GB)
→ Apple M2 Max/Ultra (64 GB)

90B model:

→ 2–4× A100 80 GB GPUs
→ Cloud / datacenter hardware

Open weight model:
meta-llama/Llama-3.2-11B-Vision-Instruct
on Hugging Face 🤗

Cannot generate images — text output only.
Single image per input (standard config).

05

Fine tunning Llama Vision 3.2

Medical Imaging · Dataset Curation · LoRA · Clinical Examples · Evaluation

Why Fine-Tune for Medical Imaging?

- Pre-trained VLMs train on internet images — NOT medical scans
- Medical images have unique characteristics: X-ray density, MRI contrast, H&E staining
- Base models lack clinical vocabulary, anatomical knowledge, and report structure
- Safety is paramount: a hallucination in medical AI can directly harm patients
- Fine-tuning on curated medical datasets bridges the distribution gap

⚠ Medical AI Caveat: Fine-tuned models must be clinically validated and used only as decision-support tools — never as standalone diagnostic systems. Regulatory approval (FDA, CE Mark) required for clinical deployment.

✗ Base Model (no medical FT)

Prompt: "Describe this chest X-ray."

Output: "The image shows a black and white photograph of a human torso with some brighter and darker regions."

✅ After Medical Fine-Tuning

Prompt: "Describe this chest X-ray."

Output: "PA chest radiograph demonstrates clear lung fields bilaterally. Cardiac silhouette is within normal limits. No pleural effusion, pneumothorax, or consolidation. Bony structures intact."

Medical Imaging Modalities for VLM Fine-Tuning

Chest X-Ray (CXR)

Most common modality. Detects pneumonia, cardiomegaly, pneumothorax. CheXpert: 224K images.

CT Scans

3D cross-sections. Lung nodules, organ abnormalities, trauma. Requires slice-by-slice or 3D approach.

MRI

Soft tissue: brain tumours, joint injuries, spinal cord. Multiple sequences: T1, T2, FLAIR, DWI.

Histopathology (H&E)

Stained tissue slides. Cancer grading, cell classification. TCGA: 11,000+ whole-slide images.

Fundus Imaging

Retinal photographs. Diabetic retinopathy, glaucoma screening. Kaggle DR: 35K labelled images.

Ultrasound

Real-time soft tissue, cardiac, obstetric imaging. More artefact-prone; requires specialist interpretation.

Dataset Preparation for Medical VLM Fine-Tuning

1. **Data Sourcing**
   CheXpert, MIMIC-CXR, PadChest (CXR); PathVQA, VQA-RAD (Medical QA)
2. **Report Pairing**
   Link each image to its free-text radiology report → (image, text) pairs
3. **Report Segmentation**
   Split reports into structured sections: Findings, Impression, Recommendations
4. **QA Generation**
   Use LLM to convert reports into instruction-following format: Question → Answer
5. **De-identification**
   Remove all PHI (patient names, IDs, DOB, dates) — HIPAA/GDPR compliance
6. **Quality Filtering**
   Remove corrupted images, low-quality reports, duplicate studies; expert review

Original MIMIC-CXR Report

```
FINDINGS: The lungs are clear bilaterally. No
pleural effusion. Cardiac silhouette is
normal. No pneumothorax.
IMPRESSION: Normal chest radiograph.
```

Converted to instruction format:

```
{"image": "cxr_00123.jpg",
"question": "Is there pneumothorax?",
"answer": "No. The radiograph shows
no evidence of pneumothorax.
Lung fields are clear bilaterally
with normal pleural margins."}
```

Medical Training Datasets: Curation and Quality

- CheXpert (Irvin 2019, Stanford): 224,316 chest X-rays from 65,240 patients; 14 radiological conditions; labels extracted by NLP from radiology reports — some label noise present
- MIMIC-CXR (Johnson 2019, MIT/Beth Israel): 227,827 chest X-rays with 227,835 free-text radiology reports; de-identified; linked to MIMIC-III EHR data — gold standard for report generation tasks
- TCGA (The Cancer Genome Atlas): 11,000+ H&E whole-slide images across 33 cancer types; genomic, clinical, and survival data linked — enables multi-modal molecular prediction
- UK Biobank Imaging (2019–present): 100,000+ participants with brain MRI, heart MRI, DEXA, retinal OCT — largest population imaging study; enables disease risk prediction from imaging phenotypes
- RSNA Challenges (2018–2024): annual AI radiology competitions providing pneumonia, bone age, spinal pathology, and brain tumour detection datasets with expert annotations
- Quality issues: label noise (NLP-extracted vs expert-annotated), selection bias (academic medical centres), scanner/protocol heterogeneity, class imbalance — all must be addressed in fine-tuning pipeline

Data Quality Checklist

- ✅ Expert annotation vs NLP-extracted labels? (prefer expert)
- ✅ Multi-site diversity (scanner, geography, demographics)?
- ✅ De-identification (HIPAA PHI removal verified)?
- ✅ Class balance or weighted sampling strategy?
- ✅ Train/val/test split — no patient overlap?
- ✅ Licensing: MIMIC requires PhysioNet credentialing
- ✅ QA: image quality filtering (motion artefacts, underexposure)?
- ✅ Annotation guideline versioning — consistent labelling protocol?

Fine-Tuning Llama 3.2 Vision on Medical Data

Recommended Strategy: LoRA (Low-Rank Adaptation) on Cross-Attention Layers

Full Fine-Tuning

+ Best performance
− Needs 40–80 GB VRAM for 11B; expensive

⭐ LoRA (recommended)

+ 90%+ less memory; freezes base model
− Slight gap vs full FT

Freeze ViT / Train LLM

+ Preserves visual representations
− Loses vision-language alignment

Freeze LLM / Train ViT+Adapter

+ Preserves text capabilities
− Limited medical vocabulary adaptation

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LoRA Configuration (Hugging Face PEFT)

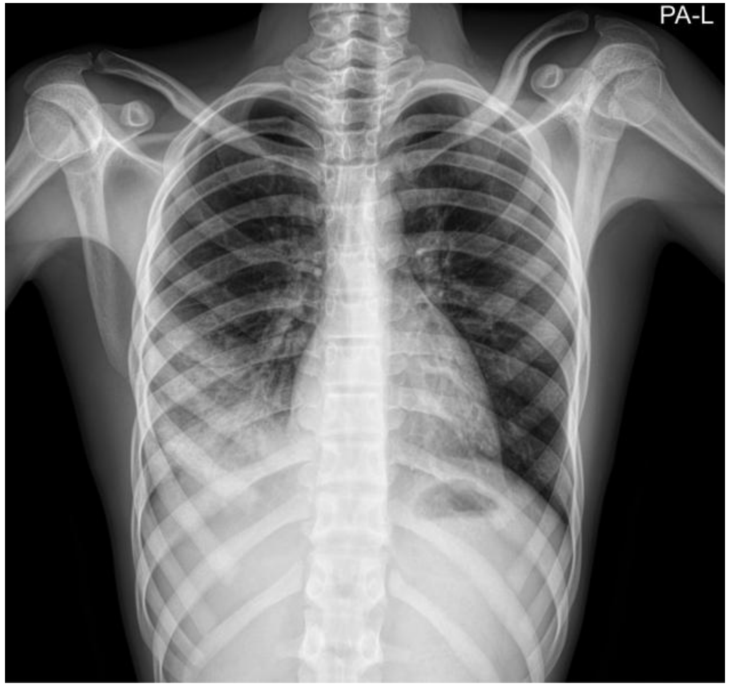
```
from peft import LoraConfig, get_peft_model

lora_cfg = LoraConfig(
  r=16,         # rank
  lora_alpha=32,
  target_modules=[
    "q_proj","k_proj",
    "v_proj","o_proj"
  ],
  lora_dropout=0.05,
  bias="none",
  task_type="CAUSAL_LM"
)
model = get_peft_model(model, lora_cfg)
```

Training Config

LR: 2e-5 | Batch: 8–16 | Epochs: 3–5
Optimiser: AdamW | BF16 precision
Warmup: 200 steps | Grad checkpointing: on

Healthcare Example: Chest X-Ray Analysis



Prompt: "Describe the findings and provide a differential diagnosis."

✅ **Fine-Tuned Model Output**

"FINDINGS: PA chest radiograph demonstrates increased airspace opacity in the right lower lobe consistent with consolidation. Air bronchograms visible within the opacity. Left lung clear. Cardiac silhouette within normal limits. No pleural effusion. Costophrenic angles sharp bilaterally.

IMPRESSION: Right lower lobe consolidation.

DIFFERENTIAL DIAGNOSIS:
- Community-acquired pneumonia (most likely)
- Aspiration pneumonitis
- Atelectasis (less likely)
- Pulmonary infarct (less likely)

RECOMMENDATION: Correlate with clinical symptoms, temperature, CRP. Consider antibiotic therapy if infectious aetiology confirmed."

Note: model outputs structured Findings / Impression / Differential / Recommendation format learned from radiology reports.

Chest X-Ray Analysis: Normal vs Abnormal Patterns

PA Chest X-Ray — Normal Reference Anatomy

Q: What are the key anatomical landmarks on a normal PA chest X-ray?

Fine-Tuned Model Output:

Output: 'Normal PA chest radiograph demonstrates:
Lung fields: clear bilaterally with visible vascular markings to the periphery — no consolidation, effusion, or pneumothorax. Cardiac silhouette: cardiothoracic ratio <0.5 within normal limits. Hila: bilaterally symmetric with normal hilar angles (right>left). Mediastinum: normal width (<8cm on PA), no tracheal deviation. Diaphragm: right hemidiaphragm higher than left — normal variant. Costophrenic angles: sharp bilaterally — no blunting (effusion). Bony structures: no rib fractures, no lytic lesions noted.'

Pathological CXR — Pneumothorax

Q: Are there signs of pneumothorax?

Model Output:

'Right-sided pneumothorax detected.
Visible pleural line at the right apex, approximately 2.5cm from the chest wall.
No shift of the mediastinum — likely small-to-moderate pneumothorax.
Left lung unremarkable.
IMPRESSION: Right pneumothorax.
RECOMMENDATION: Urgent clinical review; consider intercostal drain if respiratory compromise.'

Key teaching: the AI model learns systematic review of anatomical regions — ensuring no structure is overlooked. The model's structured output mirrors the Chest X-Ray reading protocol taught in radiology training.

CT Scan: Lung Nodule Detection and Characterisation

🔬 CT Chest — Axial Slice (Lung Window)

Prompt: 'Identify any pulmonary nodules and characterise them using Fleischner criteria.'

✅ Fine-Tuned Model Output:

Output: 'Axial CT chest in lung window setting reveals: RIGHT UPPER LOBE: 8mm solid pulmonary nodule at the right upper lobe posterior segment (Series 3, Image 47). Morphology: solid, non-calcified, irregular margin. No satellite nodules identified. LEFT LUNG: No nodules or masses identified. Additional findings: mild emphysematous changes bilateral apices. FLEISCHNER RECOMMENDATION (solid nodule, 8mm, low-risk): CT follow-up at 3 months to assess for growth/stability. DIFFERENTIAL: Primary lung carcinoma (excluded if stable at 3m), focal scar, granuloma (if calcified on HRCT), carcinoid.'

🔬 CT Abdomen — Liver Lesion

Prompt: 'Characterise the hepatic lesion on this portal venous phase CT.'

✅ Model Output:

'Hypodense lesion segment 6 liver, 2.1cm. On portal venous phase: peripheral nodular enhancement with centripetal fill-in pattern. No capsular retraction. IMPRESSION: Hepatic haemangioma (most likely). Enhancement pattern classic. No features of HCC or metastasis. 6-month MRI follow-up or contrast CEUS for confirmation.'

Clinical significance: the 8mm threshold triggers Fleischner Society guideline-based follow-up recommendations. The model correctly applies clinical decision rules from the training corpus — demonstrating learned clinical reasoning, not just pattern matching.

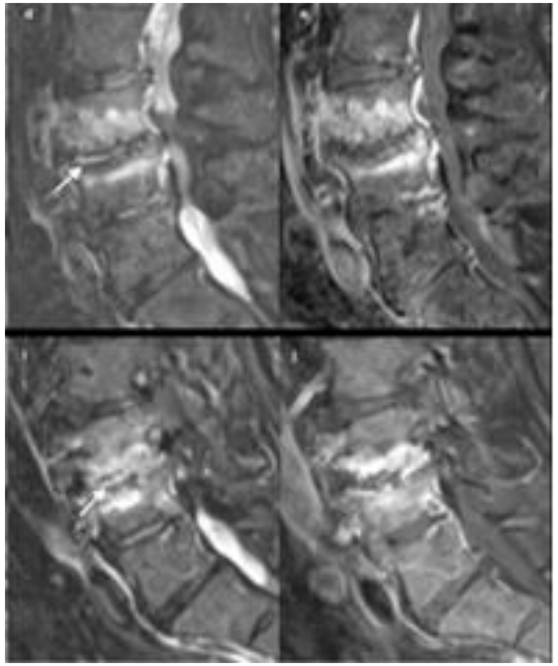
MRI Spine: Disc Disease and Musculoskeletal Assessment

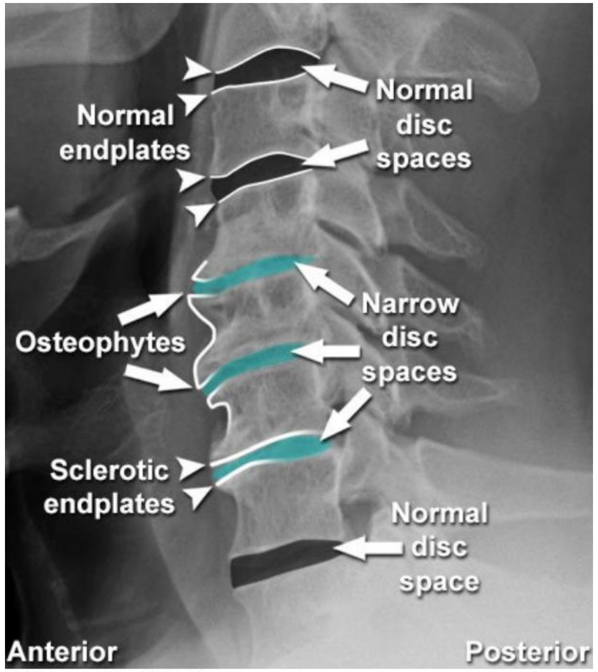
🦴 **T2 Sagittal Lumbar Spine MRI**

Prompt: 'Grade the lumbar disc disease and identify any neural compromise.'

✅ **Fine-Tuned Model Output:**

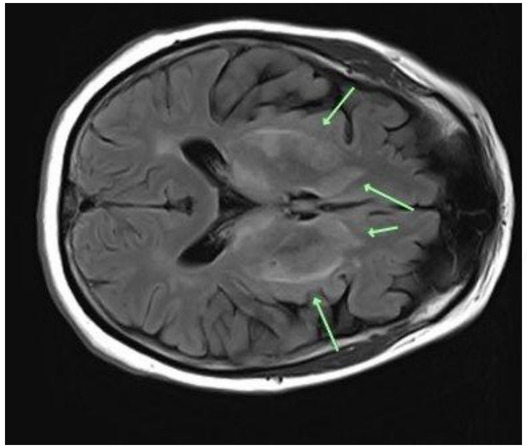
Output: 'T2 SAGITTAL LUMBAR MRI SYSTEMATIC REVIEW:
L1/L2: Normal disc height and signal. No disc protrusion.
L2/L3: Mild disc dehydration (Pfirrmann Grade 2). No herniation.
L3/L4: Moderate disc degeneration (Pfirrmann Grade 3).
Mild posterior disc bulge — no significant canal stenosis.
L4/L5: Severe disc degeneration (Pfirrmann Grade 4).
Right paracentral disc protrusion 6mm with mild right
L5 nerve root compression. Foraminal narrowing right.
L5/S1: Collapsed disc space, end-plate changes (Modic Type 2).
Broad-based disc herniation with bilateral S1 nerve
root contact. Severe bilateral foraminal stenosis.

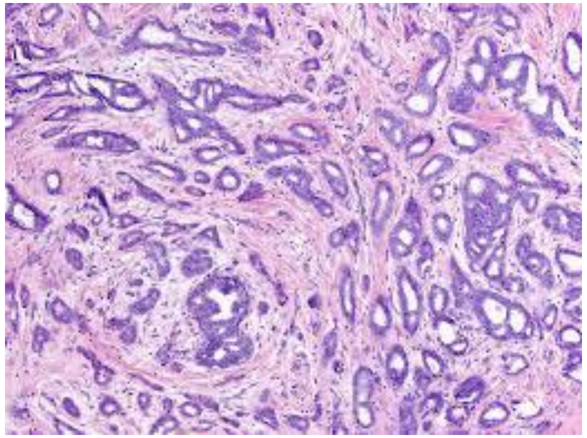




RSNA AI Challenge 2020 focused on lumbar spine MRI grading. Pfirrmann grading classification achieves >85% agreement with expert radiologists. VLMs add the advantage of free-text disc-by-disc reporting matching clinical radiology report format.

Healthcare Examples: MRI Brain and Histopathology





Q: What abnormalities are visible on this T2-FLAIR MRI?

✅ "T2-FLAIR MRI demonstrates multiple periventricular and subcortical white matter hyperintensities bilaterally. Findings are consistent with moderate white matter disease. Differential includes chronic small vessel ischaemia or demyelinating disease (MS — assess McDonald criteria). CSF oligoclonal bands and MR spectroscopy recommended for further characterisation."

Q: Grade and classify this tumour.

✅ "H&E section shows infiltrating ductal carcinoma (IDC), breast. Marked nuclear pleomorphism, prominent nucleoli, high mitotic index (>20/10 HPF). Tubule formation <10%. Nottingham Grade: 3+3+3 = Score 9 → Grade 3 (poorly differentiated). Lymphovascular invasion noted. Recommend ER/PR/HER2 IHC and Ki-67 proliferation index."

MRI Brain: Tumour Characterisation and Differential Diagnosis

🧠 **T1 Post-Contrast + T2-FLAIR MRI Brain**

Prompt: 'Analyse this MRI for a space-occupying lesion and provide a structured differential diagnosis.'

✅ **Fine-Tuned Model Output:**

Output: 'MULTISEQUENCE BRAIN MRI ANALYSIS:
T1 post-contrast: Ring-enhancing mass, left frontal lobe,
3.2×2.8cm. Thick irregular wall enhancement.
Central non-enhancing core (necrosis).
T2-FLAIR: Extensive surrounding vasogenic oedema
with mass effect — 6mm rightward midline shift.
DWI: Restricted diffusion within core — high cellularity.
MRS (if available): Elevated choline, reduced NAA.
DIFFERENTIAL DIAGNOSIS (by probability):
1. High-grade glioma (GBM) — most likely: ring enhancement, necrosis, oedema
2. Brain metastasis — single lesion, frontal location, age >50
3. CNS lymphoma — if immunocompromised: may show restricted diffusion
RECOMMENDATION: Neurosurgery referral; tissue biopsy for histopathology,
MGMT methylation, IDH mutation status. MR spectroscopy for metabolic profiling.'

The model demonstrates multi-sequence integration: T1+C, T2-FLAIR, DWI — correlating findings across sequences the way a neuroradiologist would. Training on radiology teaching files and annotated MRI reports enables this structured differential generation.

Histopathology: Cancer Grading and Molecular Prediction

🔬 H&E Breast — IDC, Nottingham Grading

Q: Grade this breast cancer specimen using the Nottingham grading system.

✅ **Fine-Tuned Model Output:**

Output: 'H&E section, breast core biopsy:
HISTOLOGICAL FEATURES:
  Tubule formation: <10% of tumour area — Score 3
  Nuclear pleomorphism: marked, large irregular nuclei — Score 3
  Mitotic count: 22 per 10 HPF (0.55mm² field) — Score 3
NOTTINGHAM GRADE: 3+3+3 = Score 9 → Grade 3 (poorly differentiated)
ADDITIONAL FEATURES: Lymphovascular invasion present.
  Peritumoural inflammatory infiltrate: moderate.
IHC RECOMMENDATIONS: ER, PR, HER2 (FISH if IHC 2+), Ki-67.
MOLECULAR SUBTYPE PREDICTION: HER2-enriched or Triple-negative
  (given Grade 3, high mitotic index — pending IHC confirmation).
PROGNOSIS: Grade 3 IDC associated with higher recurrence risk;
  chemotherapy benefit to be assessed using Oncotype DX score.'

🔬 H&E Colorectal — MSI Prediction

Q: Predict microsatellite instability status from H&E.

✅ **Model Output:**

MSI prediction from H&E morphology:
FEATURES SUPPORTING MSI-H:
  Tumour-infiltrating lymphocytes (TILs):
  high density (>2/HPF) — present ✓
  Crohn-like lymphoid reaction — present ✓
  Poorly differentiated (mucinous) — absent
MODEL PREDICTION: MSI-H probability 0.78
(threshold 0.5 → classified MSI-H)
RECOMMENDATION: IHC for MLH1/MSH2/
MSH6/PMS2. PCR fragment analysis
for confirmation. MSI-H predicts
benefit from immunotherapy (PD-1).

AI pathology systems must learn the Nottingham/Elston-Ellis criteria from annotated WSI (whole-slide images). The model is trained on TCGA dataset annotations and expert-graded teaching cases from the College of American Pathologists.

Whole-Slide Imaging: Multi-Instance Learning for Pathology

- Challenge: WSI (whole-slide images) are gigapixel images (100,000×100,000 pixels) — too large to process as a single input to any VLM
- Multi-Instance Learning (MIL): divide WSI into thousands of small tiles (256×256 at 20× magnification); aggregate tile features to make slide-level predictions
- VLM integration: CONCH and PLIP models use CLIP-style contrastive training on pathology image-text pairs from pathology textbooks and PubMed figure captions
- Applications: tumour subtype classification (NSCLC: adenocarcinoma vs squamous cell), grade prediction, MSI status, molecular marker prediction (BRAF, KRAS mutations)
- Performance: TCGA benchmark — VLM-based WSI analysis achieves AUC >0.95 for lung cancer subtyping, approaching pathologist performance on many tasks

MIL Architecture

- WSI tiling: ~10,000 tiles per slide at 20× magnification
- Feature extraction: pre-trained ViT or ResNet encoder per tile
- Attention mechanism: learn weights α_i for each tile
- Slide embedding: $z = \Sigma \alpha_i \times f_i$ (weighted sum of tile features)
- Classifier: linear head on slide embedding → subtype label
- Heatmap: visualise α_i as colour overlay on WSI
- Training: slide-level labels only — no per-tile annotations needed

Ultrasound: Cardiac and Obstetric AI Applications

🫀 Echocardiography — Parasternal Long Axis

Prompt: 'Assess left ventricular function from this echo image.'

✅ Fine-Tuned Model Output:

Output: 'Parasternal long-axis view. LV cavity mildly dilated (LVEDD ~6.1 cm). Wall motion: global hypokinesia, anterior wall most affected. Estimated LVEF: 35–40% (mildly-severely reduced) by visual assessment. Mitral valve: posterior leaflet restricted motion. Aortic root: 3.2cm, normal. No pericardial effusion. IMPRESSION: Dilated cardiomyopathy with moderately reduced LVEF ~38%. Recommend formal biplane Simpson LVEF measurement. Cardiology referral for HF management.'

🤰 Obstetric Ultrasound — Fetal Biometry

Prompt: 'Measure standard fetal biometry planes.'

✅ Model Output:

BPD (biparietal diameter): 72mm
HC (head circumference): 261mm
AC (abdominal circumference): 248mm
FL (femur length): 53mm
EFW (Hadlock formula): 1,410g ±15%
Gestational age estimate: 29+3 weeks
Percentile: AC 38th centile — appropriate for gestational age. AFI: 14cm — normal.
IMPRESSION: Appropriately grown fetus.
Next scan: growth scan at 34 weeks.

Echocardiography AI: models trained on the EchoNet-Dynamic dataset (10K echo videos, LVEF labels from Stanford) achieve EF estimation comparable to cardiologists (MAE ~4%). Video-capable VLMs extend this to multi-view assessment.

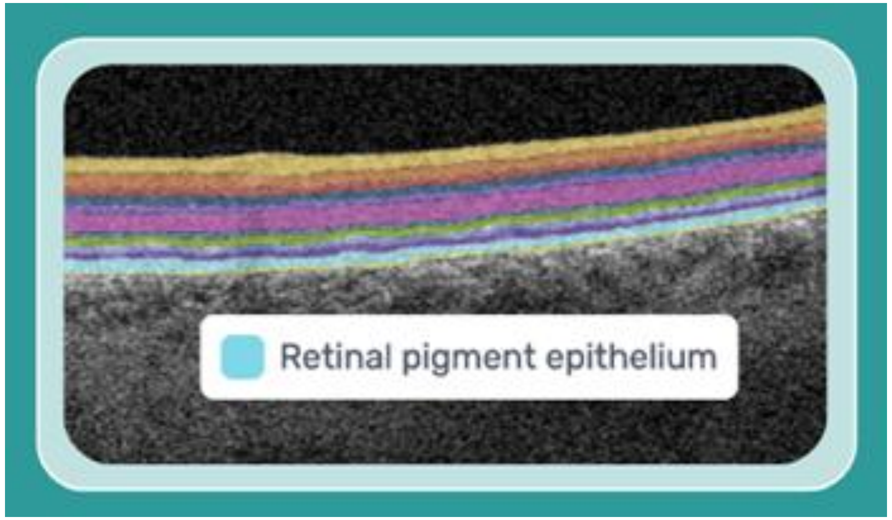
Optical Coherence Tomography: Retinal Layer Analysis

👁️ OCT Macular Cross-Section

Prompt: 'Analyse retinal layer integrity on this OCT scan and identify any pathology.'

✅ Fine-Tuned Model Output:

Output: 'MACULAR OCT ANALYSIS:
Foveal pit: normal contour preserved.
DRUSEN: Multiple medium drusen (63–124μm) under RPE layer, predominantly temporal fovea.
RPE layer: intact centrally, subtle attenuation temporal to fovea.
Ellipsoid zone (EZ): intact centrally.
Subretinal fluid: absent.
Intraretinal fluid (IRF): absent.
CLASSIFICATION: Age-related macular degeneration, intermediate AMD (AREDS category 3).
RISK: ~18% chance of progression to late AMD within 5 years (AREDS2 model).



OCT produces ~65,000 cross-sectional retinal images per second. AI systems trained on the UK Biobank Imaging Study (100K OCT scans) achieve expert-level AMD staging. Drusen segmentation enables sub-RPE volume quantification for longitudinal monitoring.

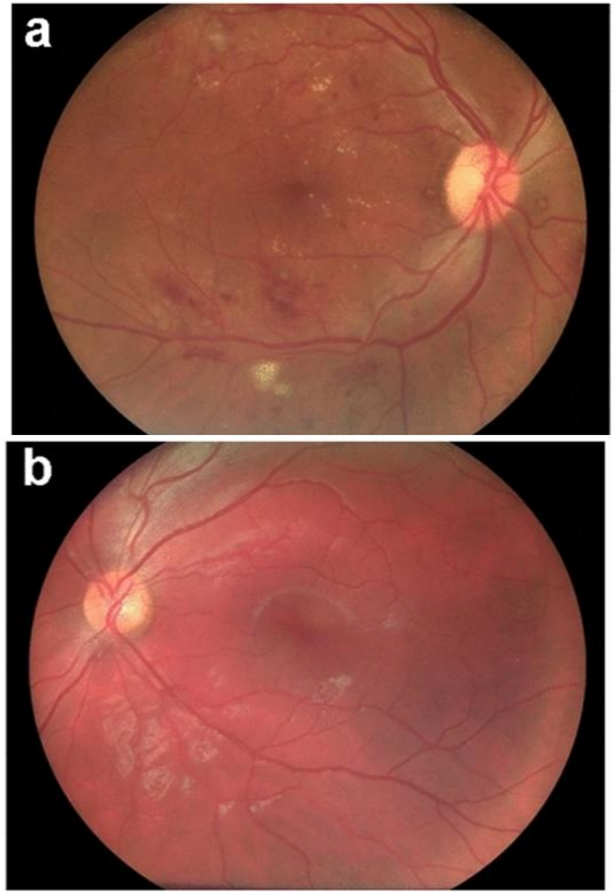
Fundus Imaging: Diabetic Retinopathy Grading and Screening

👁️ **Fundus Photograph — Right Eye**

Prompt: 'Grade this fundus photograph for diabetic retinopathy severity.'

✅ **Fine-Tuned Model Output:**

Output: 'FUNDUS PHOTOGRAPH ANALYSIS — RIGHT EYE:
MICROANEURYSMS: Multiple dot haemorrhages temporal to fovea (>5 in nasal quadrant).
HARD EXUDATES: Partial circinate ring superior to fovea — threatening macular oedema.
SOFT EXUDATES: One cotton-wool spot superior arcade — indicates retinal ischaemia.
HAEMORRHAGES: Flame-shaped superior and dot/blot inferotemporal.
NVD/NVE: No neovascularisation of disc or elsewhere identified.
GRADING: Moderately severe non-proliferative diabetic retinopathy (NPDR)
ETDRS Level 47 — High-risk features present (>20 intraretinal haemorrhages in ≥4 quadrants).
CLINICALLY SIGNIFICANT MACULAR OEDEMA (CSMO): Possible — OCT recommended.



From: [Artificial intelligence in diabetic retinopathy screening: clinical assessment using handheld fundus camera in a real-life setting]

Diabetic retinopathy screening is one of the most validated AI applications in medicine. Google's EyePAT and similar systems achieve AUC >0.99 for referable DR. Fine-tuned Llama 3.2 on Kaggle DR (35K images) and EyePACS dataset enables structured ETDRS grading with clinical recommendations.

Proton & Photon Radiotherapy: AI-Assisted Dose Planning

- Clinical challenge: precise dose delivery to tumour while sparing surrounding organs-at-risk (OAR) — brain, spinal cord, bowel, heart
- Role of multimodal AI: VLMs can interpret CT/MRI planning scans, identify target volumes (GTV/CTV/PTV), and auto-segment OARs
- DNA repair relevance: accurate dose prediction requires modelling how DNA double-strand breaks are repaired — anomalous subdiffusive motion of DSB ends during NHEJ is key to understanding radiobiological response
- Proton Bragg peak advantage: protons deposit maximum dose at a precise depth — VLMs can optimise beam angles by reasoning about 3D anatomy from multi-planar CT reconstructions
- Clinical impact: reduced treatment planning time from hours to minutes; more consistent target coverage; fewer normal tissue complications

VLM Applications in Radiotherapy

- Auto-segmentation: VLM identifies GTV, CTV, PTV and OARs on planning CT
- Plan QA: model reviews dose-volume histograms and flags deviations from protocol
- Adaptive RT: re-plan detection when anatomy changes mid-course (weight loss, tumour response)
- Report generation: structured clinical summary of dose plan for physics/oncology review
- Outcome prediction: correlate dose distribution with toxicity outcomes from historical cases
- Research link: anomalous diffusion models of DNA repair directly inform LET-RBE relationships

Evaluation and Safety in Medical VLMs

Evaluation Metrics

RadGraph F1
Compares clinical entities/relations between output and reference report

BERTScore (ClinicalBERT)
Semantic similarity using a domain-adapted BERT model

Expert Radiologist Rating
Clinicians score on accuracy, completeness, actionability — the gold standard

Finding-level Sens / Spec
Per-condition: sensitivity and specificity for pneumonia, cardiomegaly, etc.

Hallucination Rate
Fraction of outputs containing clinically incorrect or fabricated findings

⛔ Safety Requirements

- Never fabricate diagnoses with false confidence
- All outputs must include uncertainty language
- Test on diverse demographics to prevent bias
- Human clinician review mandatory for patient-facing use
- Regulatory: FDA 510(k) / EU MDR / CE Mark required

✅ Best Practices

- Use structured output: Findings / Impression / Differential
- Provide calibrated uncertainty ("likely", "possible", "exclude")
- Log all outputs for audit and continuous monitoring
- Combine with retrieval-augmented generation for evidence

Zero-Shot vs Fine-Tuned VLMs on Medical Imaging Tasks

Task / Dataset	Metric	Base Llama 3.2 (Zero-Shot)	Med Fine-Tuned Llama 3.2	Expert Radiologist
CheXpert CXR — 14 findings	AUC (mean)	0.62	0.87	0.90
MIMIC-CXR Report Gen	RadGraph F1	0.21	0.61	0.85 (inter-rater)
PathVQA — Pathology	Acc (closed)	0.41	0.73	—
VQA-RAD — Clinical VQA	Acc (open)	0.38	0.67	—
PadChest Pneumonia Det.	Sensitivity	0.48	0.83	0.91
Kaggle DR — 5-class grade	Kappa (quadratic)	0.29	0.71	0.78–0.82
HAM10000 Dermoscopy	Top-3 Acc	0.55	0.84	0.91

Fine-tuning dramatically reduces the gap between general VLM and expert clinician. The remaining gap is mostly in borderline/ambiguous cases.
Data sourced from published benchmarks (2023–2024). Results for Llama 3.2 fine-tuning are indicative based on comparable architecture studies.

Hallucination in Medical AI: Real-World Examples and Consequences

Object Hallucination

- Model reports 'bilateral pleural effusion' on a chest X-ray that shows only minor blunting of the left costophrenic angle
- Fabricated finding could lead to unnecessary thoracocentesis procedure
- Root cause: training data imbalance — effusion is common, model over-predicts
- Mitigation: calibrated confidence scores, output verification, dual-reader protocols
- Test: HallusionBench — medical subset measures fabricated clinical findings rate

Attribute Error

- Model grades a prostate biopsy as Gleason 4+4=8 when pathologist scores 3+3=6 — a clinically critical difference
- Grade 8 triggers immediate aggressive treatment; Grade 6 may only require active surveillance
- Root cause: subtle morphological differences at the borderline — ambiguous ground truth
- Mitigation: report uncertainty: 'Gleason 3+4 vs 4+3 — equivocal; recommend expert second opinion'
- Key: AI must communicate calibrated uncertainty, not false precision

Fabricated Evidence

- Model references 'air bronchograms in the left lower lobe' when no consolidation exists — the feature it describes is not present
- Plausible hallucination: air bronchograms are real and common — model has learned they co-occur with consolidation
- Clinical risk: radiologist may defer to AI report without re-reading the scan if time-pressured
- Mitigation: attention map/saliency overlays showing which image regions drove the output
- Regulation: FDA requires hallucination rate disclosure and human oversight for clinical AI decisions

Clinical Validation: From Research to Deployment

1

Technical validation

Benchmark performance on held-out test sets (RadGraph F1, BERTScore-Clinical, VQA accuracy, sensitivity/specificity per finding). Must exceed published baseline and demonstrate statistical significance. Report performance disaggregated by patient demographics, scanner type, and institution.

2

Clinical reader study

Radiologists/pathologists read studies with and without AI assistance. Primary endpoints: time-to-diagnosis, diagnostic accuracy, inter-reader agreement, missed finding rate. Blinded, randomised design. Powered for non-inferiority (AI ≥ unassisted reader) or superiority depending on intended use.

3

Prospective pilot

Deploy in restricted clinical workflow at a single institution. Collect prospective data on AI outputs vs final reports. Monitor hallucination rate, time savings, workflow integration issues. Regulatory: requires ethics approval (NHS REC, IRB), informed consent, and data governance agreement.

4

Regulatory submission

For UK: NHS DTAC (Digital Technology Assessment Criteria) + UKCA/CE Mark under UK MDR 2002. For EU: CE Mark under EU MDR 2017/745 — clinical investigation data required for Class IIb/III devices. For USA: FDA 510(k) premarket notification or De Novo classification for novel AI indications.

5

Post-market surveillance

Continuous monitoring: monthly audit of AI output vs final report discrepancy rate. Model drift detection: performance degrades as scanner hardware, protocols, and patient population shift. Trigger for re-training: significant drift, new disease emergence (e.g., COVID-19 CXR patterns), scanner upgrade.

Regulatory Framework for AI Medical Devices

- EU MDR 2017/745: AI diagnostic tools are Class IIb Medical Devices — require Notified Body assessment, clinical evidence, and post-market clinical follow-up (PMCF)
- FDA: AI/ML-based software as Medical Device (SaMD). 510(k) for substantial equivalence to predicate; De Novo for novel devices. Predetermined Change Control Plan (PCCP) for continuous learning systems
- UK: Post-Brexit — UKCA mark replaces CE for GB market; separate MHRA registration required. NHS AI Lab publishes AI and Digital Regulations Service guidance
- All jurisdictions require: intended use statement, training/validation data documentation, bias and fairness assessment, human oversight mechanism, incident reporting plan
- IVD Regulation: AI for in vitro diagnostics (pathology, cytology) falls under IVDR 2017/746 in EU — stricter requirements, common specification compliance
- Key principle: AI as decision-support, never autonomous diagnostic system for high-risk decisions — a clinician must review and take responsibility for all patient-facing outputs

Timeline to Clearance

- Phase 1: Algorithm development & internal validation — 12–24 months
- Phase 2: Multi-site clinical study — 12–18 months
- Phase 3: Regulatory submission — 3–6 months preparation
- FDA 510(k): 90-day review (Breakthrough Device: accelerated)
- EU MDR: 12–24 months Notified Body review
- Total: typically 3–5 years from research to clinical use
- Example: IDx-DR (diabetic retinopathy screening) — first autonomous AI diagnostic cleared by FDA 2018

Bias and Fairness in Medical AI Models

- Representation bias: MIMIC-CXR and CheXpert are US hospital datasets — predominantly white, US healthcare system. Performance degrades for patient populations underrepresented in training (South Asian, African populations, paediatric)
- Scanner bias: model trained on GE scanner data may underperform on Siemens or Philips scanners due to subtle differences in image reconstruction, noise characteristics, and contrast curves
- Label bias: radiology reports written under time pressure contain systematic errors — AI inherits biases in the label noise, not just in the images
- Sex and age bias: certain conditions present differently by demographic (e.g., MI presentation in women); models trained without stratified evaluation may perform poorly for subgroups
- Mitigation — data level: stratified sampling, oversampling rare subgroups, multi-site federated training
- Mitigation — evaluation level: mandatory disaggregated performance reporting by sex, age, ethnicity, scanner type, institution before deployment

Clinical Impact

- Obermeyer et al. (2019 Science): commercial risk prediction tool underestimated Black patients' illness severity — same risk score meant Black patients were sicker
- Retinal AI: diagnostic accuracy for dark fundus images (more common in darker-pigmented patients) significantly lower in some systems
- Chest X-ray: pneumonia detection AUC 0.90 on NIH test set — but drops to 0.76 on external dataset (Stanford)
- Mitigation: DRO (distributionally robust optimisation), fairness constraints during fine-tuning
- Regulation: FDA proposes mandatory subgroup analysis in AI/ML-SaMD submissions

Federated Learning: Privacy-Preserving Medical AI Training

- Core problem: patient data cannot leave hospital networks due to HIPAA (USA), GDPR (EU), and data processing agreements — but model quality improves with more data
- Federated Learning (McMahan 2017): train locally, share gradients (not data). Each site trains on local data, sends model updates (not images) to a central aggregator
- FedAvg algorithm: central server averages parameter updates from N sites → global model. Iterate for multiple rounds until convergence
- Differential Privacy (DP-SGD): add calibrated Gaussian noise to gradients before sharing — prevents re-identification of individual training images from gradient updates
- Secure Aggregation: cryptographic protocol ensuring aggregator cannot see individual site updates — only the aggregate
- Medical applications: FeTS Challenge (federated tumour segmentation), Nvidia FLARE platform for hospital AI consortia, NHS AI Lab federated learning infrastructure

Trade-offs

- ✅ Privacy: patient data never leaves local site
- ✅ Regulatory: simpler data governance
- ✅ Scale: aggregate learnings from many hospitals without data sharing
- ❌ Communication overhead: many rounds of gradient exchange
- ❌ Data heterogeneity: non-IID data across sites degrades convergence
- ❌ DP noise degrades accuracy vs centralised training
- Mitigation: personalised FL (FedProx, SCAFFOLD) handles non-IID better than FedAvg

Retrieval-Augmented Generation for Clinical Evidence

- Problem: VLMs can generate plausible but outdated or fabricated clinical recommendations — dangerous in a rapidly evolving field
- RAG solution: at inference time, retrieve relevant clinical guidelines, recent literature, and prior cases; inject into context alongside the image
- Architecture: image → VLM encoder → query embedding → retrieve from vector DB (PubMed embeddings, NICE guidelines, local radiology reports) → LLM generates grounded response
- Example: chest X-ray with unusual pattern → retrieve top-5 similar cases from institutional PACS with confirmed diagnoses → LLM generates differential including retrieved matches
- Clinical benefit: outputs cite evidence ('per NICE NG28 guideline 2023, CT follow-up recommended for nodules ≥6mm in low-risk patients')
- Key databases: PubMed/MEDLINE (30M articles), NICE guidelines, BNF, RCR radiology reporting guidelines, institutional report archives

RAG Architecture

- 1. Image → ViT encoder → visual embedding
- 2. Prompt → text encoder → query embedding
- 3. Fused query → cosine search in vector DB
- 4. Top-k clinical documents retrieved
- 5. Context: [image features] + [retrieved text] + [question]
- 6. LLM generates grounded, citable response
- Evaluation: citation accuracy, hallucination rate vs non-RAG
- RAG reduces hallucination by 40–60% in clinical settings (Zakka 2024)

Confidence Calibration and Uncertainty Quantification in Medical AI

- Miscalibration problem: fine-tuned models often express high confidence on ambiguous or out-of-distribution images — the model 'doesn't know what it doesn't know'
- Calibration definition: a model is well-calibrated if its stated confidence (0–1) equals its empirical accuracy over many predictions — P(correct | confidence=0.8) = 0.80
- ECE (Expected Calibration Error): measure binned confidence vs accuracy gap. Well-calibrated model has ECE < 0.05
- Post-hoc calibration — Temperature Scaling: divide logits by temperature T before softmax. T>1 softens distribution, reducing overconfidence
- Bayesian approaches: MC Dropout (Gal & Ghahramani 2016) — run forward pass N times with dropout active; variance of predictions = uncertainty
- Linguistic calibration: prompt model to express uncertainty using hedging language: 'likely', 'possible', 'cannot exclude', 'recommend further investigation'

Clinical Uncertainty Language

- High confidence: 'This is consistent with lobar pneumonia'
- Moderate confidence: 'Findings likely represent early consolidation'
- Low confidence: 'Cannot exclude early infiltrate — recommend repeat in 24h'
- Ambiguous: 'Equivocal for effusion — correlate with clinical presentation and lateral film'
- Out-of-distribution: 'Image quality insufficient for confident assessment'
- Key: model should NEVER output unqualified diagnosis — always hedge appropriately
- Evaluation: compare model uncertainty vs radiologist disagreement rate as ground truth

Future Directions: Foundation Models for Healthcare

- Med-Gemini (2024): Google's multimodal model trained on medical literature, EHR, imaging, genomics — approaching expert-level performance on USMLE, MedQA benchmarks
- 3D volumetric models: CT and MRI are inherently 3D — next-generation VLMs will process full volumes, not individual slices, enabling better lesion tracking and volumetric measurements
- Multi-modal EHR integration: combining radiology images, pathology, labs, notes, genomics in a unified representation — a true AI clinician that reasons across all patient data
- Video and surgical AI: real-time VLMs for intraoperative guidance — identifying surgical planes, instruments, anatomy during laparoscopic or robotic procedures
- Personalised medicine: VLMs trained on longitudinal patient data can model disease progression, predict treatment response, and tailor recommendations to individual patient profiles
- Agentic AI systems: VLMs as agents that can retrieve evidence, order tests, draft referral letters, and coordinate care — under clinician supervision

Challenges Ahead

- Data: high-quality annotated medical image datasets remain scarce and expensive to create
- Generalisation: models that work in academic centres often fail in community hospitals
- Trustworthiness: clinicians need explainability — not just accuracy scores
- Integration: legacy hospital IT systems are difficult to interface with AI
- Workforce: AI augments but requires radiologists/pathologists trained to work with AI effectively
- Ethics: liability, informed consent, algorithmic accountability remain unresolved
- Opportunity: AI could democratise high-quality medical imaging interpretation globally

From VLM to Clinical Decision Support System (CDSS)

- CDSS definition: a system that analyses patient data and provides actionable clinical recommendations — decision support, not autonomous decision-making
- VLM-powered CDSS architecture: EHR data (labs, notes, medications) + imaging (PACS) + VLM reasoning = integrated patient-level clinical summary
- Agentic AI (emerging): VLM as orchestrator — tool calling enables the model to query lab results, retrieve prior imaging, access drug databases, and generate a structured differential diagnosis
- Example workflow: patient admitted with dyspnoea → VLM agent pulls CXR (PACS), ECG (EHR), BNP (labs), prior echo → generates: 'Consistent with acute decompensated heart failure (LVEF 35%). Recommend IV furosemide, daily weight, fluid restriction, echo on admission.'
- Evidence-based recommendations: RAG from UpToDate, NICE guidelines, BNF ensures recommendations are current and guideline-concordant
- Human-in-the-loop: every CDSS recommendation is advisory — attending clinician retains full responsibility; AI explicitly states this in every output

Integration Levels

- Level 1: Report drafting — AI drafts, radiologist edits (deployed today)
- Level 2: Triage flagging — AI flags urgent findings for immediate review (deployed in some centres)
- Level 3: Multi-modal summary — AI integrates labs + imaging + notes (research prototype stage)
- Level 4: Agentic CDSS — AI queries multiple systems, reasons, recommends (emerging)
- Key challenge: interoperability — hospital IT systems are fragmented (different PACS, EHR, lab vendors)
- HL7 FHIR R4: standardised API enabling cross-system data access — a prerequisite for Level 3/4

Summary: Key Takeaways from This Lecture

01 Transformers are universal

The same attention mechanism drives both text and vision — images become sequences of patch tokens.

02 CLIP bridged the gap

Contrastive pre-training created a shared embedding space that powers almost every multimodal LLM built since 2021.

03 Three fusion strategies

Cross-attention (Flamingo/Llama 3.2), Q-Former (BLIP-2), and concatenation (LLaVA) offer trade-offs in efficiency and performance.

04 Llama 3.2 = open + modular

Frozen Llama 3.1 + ViT-H/14 + gated cross-attention delivers strong vision capability with zero text regression.

05 Fine-tuning enables specialisation

LoRA on cross-attention layers + curated medical data converts a general VLM into a capable clinical imaging assistant.

06 Hallucination is the core challenge

In medical contexts evaluation, safety testing, and human oversight are non-negotiable — models must express uncertainty.

Key References and Further Reading

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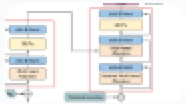
Code & Models: huggingface.co/meta-llama · github.com/haotian-liu/LLaVA · github.com/salesforce/LAVIS

Image Sources

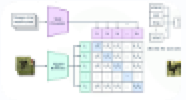


https://deeprevision.github.io/posts/001-transformer/transformer.png

Source: deeprevision.github.io

https://gowrishankar.info/blog/transformers-everywhere-patch-encoding-technique-for-vision-transformersvit-explained/patches.png

Source: gowrishankar.info



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Source: towardsdatascience.com

https://my.clevelandclinic.org/-/scassets/images/org/health/articles/10228-chest-x-ray

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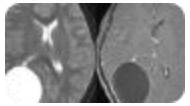

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Source: www.magonlinelibrary.com

https://upload.wikimedia.org/wikipedia/commons/b/bb/Eosinophilic%2C_basophilic%2C_chromophobic_and_amphophilic_staining.png

Source: en.wikipedia.org

Thank you for attending.

See you on the Practical today at 2 pm, LRC Annexe, room K110

Slides and lab notebooks available on the Canvas.